31/7 equals

A. 4 + 1/7

B. 4 + 3/7

C. 3 + 4/7

D. 4 + 3/31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

801

QUESTION ID

S22-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4 + 3/7

GRADING NOTE $31 = 7 \cdot 4 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 100/9 in the form q + r/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

802

QUESTION ID

S22-B1-Q5

TYPE

Short answer

CORRECT ANSWER

11 + 1/9

GRADING NOTE Form required: q + r/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 2

Multiply to check a division • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Compute $(x - 1)(x^2 + 3x - 2) + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

804

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^3 + 2x^2 - 5x + 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(x^2 + 1)(2x - 3) + (2x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

805

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$2x^3 - 3x^2 + 4x - 7$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

True or false?

$(2x + 1)(3x - 1) - 3 = 6x^2 + x - 4.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

806

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $6x^2 + x - 1 - 3.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 3)(x^2 - 3x + 5)$ equals

A. $x^3 - 4x + 15$

B. $x^3 + 15$

C. $x^3 - 4x + 5$

D. $x^3 + 6x^2 - 4x + 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

807

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^3 - 4x + 15$

GRADING NOTE $x^3 - 3x^2 + 5x + 3x^2 - 9x + 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B2-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(x - 2)(x^3 + 2x^2 + 4x + 8)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

808

QUESTION ID

S22-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$x^4 - 16$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B2-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 3

Use the Remainder Theorem to check the remainder • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Find the remainder of $x^3 + 2x^2 - 5x + 3$ divided by $x - 1$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

810

QUESTION ID
S22-B3-Q1

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

Find the remainder of $x^3 - 4x + 5$ divided by $x + 3$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

811

QUESTION ID

S22-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

True or false?

The remainder of $x^4 - 16$ divided by $x - 2$ is 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

812

QUESTION ID
S22-B3-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2^4 - 16 = 0$, so the remainder is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

The remainder of $6x^2 + x - 4$ divided by $2x + 1$ is

A. 3

B. -4

C. 1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

QUESTION ID

S22-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −3

GRADING NOTE Evaluate at $x = -1/2$: $1.5 - 0.5 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

Find the remainder of $2x^3 + x - 1$ divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

814

QUESTION ID

S22-B3-Q5

TYPE

Short answer

CORRECT ANSWER

17

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 1

Divide with unknown coefficients and higher-degree divisors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

When $x^3 + ax^2 + 2x + 4$ is divided by $x^2 - x + 1$, the remainder is a constant. Find a and the remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

816

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

a = −2; remainder 5

GRADING NOTE Form required: both. Quotient x + (a + 1); remainder (2 + a)x + (3 − a), constant when a = −2, giving 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

Rewrite $(x^4 + x^2 + 1)/(x^2 + x + 1)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

817

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - x + 1$

(remainder 0)

GRADING NOTE Form required: quotient and remainder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

True or false?

Dividing a degree-5 polynomial by a degree-2 polynomial gives a quotient of degree 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

818

QUESTION ID

S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient has degree $5 - 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The remainder of a division by $x^2 + 4$ can have degree at most

A. 0

B. 1

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

819

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1

GRADING NOTE Lower than the divisor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

Find the polynomial a(x) for which
$a(x)/(x - 2) = x^2 + 1 + 3/(x - 2)$.

a(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

820

QUESTION ID
S22-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$x^3 - 2x^2 + x + 1$

GRADING NOTE Any equivalent form. $(x - 2)(x^2 + 1) + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 2

Use division to find slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Use division to find the slant asymptote of $f(x) = (x^2 + 3x + 1)/(x + 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

822

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = x + 2$

GRADING NOTE Form required: an equation. $(x^2 + 3x + 1)/(x + 1) = x + 2 - 1/(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

Find the slant asymptote of $f(x) = (2x^2 - x - 4)/(x - 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

823

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 1

GRADING NOTE Form required: an equation. Remainder −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

True or false?

A rational function with numerator degree exactly one more than the denominator degree has a slant asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

824

QUESTION ID
S22-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The quotient is linear (unless the division is exact, when the graph is the line with a hole).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

The slant asymptote of $y = (x^2 + 1)/x$ is

A. $y = x$

B. $y = 1$

C. $x = 0$

D. $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

825

QUESTION ID
S22-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. y = x

GRADING NOTE $x + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

Explain why the graph of $y = x + 2 - 1/(x + 1)$ gets close to $y = x + 2$ for large $|x|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

826

QUESTION ID

S22-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1/(x + 1) approaches 0 as |x| grows, so y − (x + 2) approaches 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

Rewrite simple rational expressions by inspection

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Split a fraction with a sum in the numerator	**7.EE.A.1**
BACKWARD 2	Rewrite a numerator as a multiple of the denominator plus a cons…	**6.EE.A.3**
BACKWARD 3	Divide by long division when inspection fails	**HSA-APR.D.6**
FORWARD 1	Use the rewritten form to reason about values	**HSA-APR.D.6**
FORWARD 2	Graph rational functions as transformations of y = 1/x	**HSF-BF.B.3**

SET 23 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite (x + 5)/(x + 2) by inspection in the form q + r/(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

829

QUESTION ID
S23-M-Q1

TYPE
Short answer

CORRECT ANSWER
1 + 3/(x + 2)

GRADING NOTE Form required: q + r/(x + 2). x + 5 = (x + 2) + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $(2x - 1)/(x - 3)$ by inspection in the form $q + r/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

830

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

$2 + 5/(x - 3)$

GRADING NOTE Form required. $2x - 1 = 2(x - 3) + 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $(x^2 + 1)/x$ by inspection in the form $q(x) + r/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

831

QUESTION ID
S23-M-Q3

TYPE
Short answer

CORRECT ANSWER
x + 1/x

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

Rewrite (3x + 7)/(x + 1) by inspection in the form q + r/(x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

832

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 + 4/(x + 1)

GRADING NOTE Form required. 3x + 7 = 3(x + 1) + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $x^2/(x - 1)$ by inspection in the form $q(x) + r/(x - 1)$. (Hint: $x^2 = (x^2 - 1) + 1$.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

833

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x + 1 + 1/(x - 1)$

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 1

Split a fraction with a sum in the numerator • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write (x + 3)/x as a sum of two fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

835

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x/x + 3/x = 1 + 3/x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (2(x − 3) + 5)/(x − 3) as a sum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

836

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 5/(x − 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a + b)/c = a/c + b/c.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

837

QUESTION ID

S23-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division distributes over a sum in the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

$(x^2 + 1)/x$ equals

A. $x + 1/x$

B. $1 + 1/x$

C. $x + 1$

D. $x^2 + 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

838

QUESTION ID

S23-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x + 1/x$

GRADING NOTE $x^2/x + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

True or false?

c/(a + b) = c/a + c/b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

839

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Splitting works only for the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 2

Rewrite a numerator as a multiple of the denominator plus a constant • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that x + 5 = (x + 2) + c.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

841

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $2x - 1 = 2(x - 3) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

842

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

True or false?

$3x + 7 = 3(x + 1) + 7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

843

QUESTION ID

S23-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3(x + 1) + 7 = 3x + 10$; the constant is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

$x^2 - 1$ equals

A. $x(x - 1)$

B. $(x + 1)^2$

C. $(x - 1)^2$

D. $(x - 1)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

844

QUESTION ID

S23-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $(x - 1)(x + 1)$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find c so that $4x + 1 = 4(x - 2) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

845

QUESTION ID

S23-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 3

Divide by long division when inspection fails • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Divide x^2 by $x - 1$ with long division. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

847

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Quotient x + 1, remainder 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-B3-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Divide 3x + 7 by x + 1. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

848

QUESTION ID

S23-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Quotient 3, remainder 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

True or false?

Long division and inspection give the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

849

QUESTION ID

S23-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The form q + r/b is unique.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The remainder of $(2x - 1) \div (x - 3)$ is

A. -5

B. 2

C. -1

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

850

QUESTION ID

S23-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE $2 \cdot 3 - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

Divide x + 5 by x + 2. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

851

QUESTION ID

S23-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Quotient 1, remainder 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-B3-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 1

Use the rewritten form to reason about values • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

For which integers x is (x + 5)/(x + 2) an integer? Use 1 + 3/(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

853

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = −5, −3, −1, 1

GRADING NOTE Form required: all four. x + 2 must divide 3: ±1, ±3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

Explain why (2x − 1)/(x − 3) can never equal 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

854

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

It equals 2 + 5/(x − 3), and 5/(x − 3) is never 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

$(3x + 7)/(x + 1) > 3$ for every $x > -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

855

QUESTION ID

S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4/(x + 1) > 0$ when $x > -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

As x grows large, (x + 5)/(x + 2) approaches

A. 3

B. 0

C. 5/2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

856

QUESTION ID

S23-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE 3/(x + 2) → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For $x > 0$, what is the least value of $x + 1/x$, and where does it occur?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

857

QUESTION ID

S23-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2, at $x = 1$

GRADING NOTE Form required: value and location. $x + 1/x - 2 = (x - 1)^2/x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 2

Graph rational functions as transformations of y = 1/x • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Describe how the graph of $y = 1 + 3/(x + 2)$ is obtained from $y = 3/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

859

QUESTION ID
S23-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Shift left 2 and up 1.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptotes of $y = 2 + 5/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

860

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 and y = 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

y = (3x + 7)/(x + 1) has horizontal asymptote y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

861

QUESTION ID

S23-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (3x + 7)/(x + 1) = 3 + 4/(x + 1), so the asymptote is y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

$y = (x + 5)/(x + 2)$ is a translation of $y = 3/x$ by

A. left 2, up 1

B. right 2, up 1

C. left 2, up 5

D. left 5, up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

862

QUESTION ID

S23-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. left 2, up 1

GRADING NOTE $1 + 3/(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

Write (4x − 3)/(x − 1) as a transformation of k/x and give its asymptotes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

863

QUESTION ID

S23-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4 + 1/(x − 1); x = 1 and y = 4

GRADING NOTE Form required: form and both asymptotes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

Multiply and divide rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Multiply and divide fractions	**6.NS.A.1**
BACKWARD 2	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 3	Division by zero is undefined	**7.NS.A.2.b**
FORWARD 1	Simplify complex fractions	**HSA-APR.D.7**
FORWARD 2	Explain closure of rational expressions	**HSA-APR.D.7**

SET 24 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x^2 - 9)/(x^2 + x - 6) \cdot (x - 2)/(x - 3)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

866

QUESTION ID
S24-M-Q1

TYPE
Short answer

CORRECT ANSWER
1; $x \neq -3, 2, 3$

GRADING NOTE Form required: simplified result and all exclusions from the original denominators. $(x - 3)(x + 3)/((x + 3)(x - 2)) \cdot (x - 2)/(x - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

Simplify $(x + 2)/(3x^2) \div (x^2 - 4)/x$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

867

QUESTION ID
S24-M-Q2

TYPE
Short answer

CORRECT ANSWER
$1/(3x(x - 2))$; $x \neq -2, 0, 2$

GRADING NOTE Form required: result and all exclusions, including the zeros of the divisor $x^2 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

Simplify $(2x - 6)/(x + 1) \cdot (x + 1)/(x^2 - 9)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

868

QUESTION ID
S24-M-Q3

TYPE
Short answer

CORRECT ANSWER
$2/(x + 3)$; $x \neq -3, -1, 3$

GRADING NOTE Form required: result and all exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q3**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

Simplify (3x + 3)/(x − 4) ÷ (x + 1)/(x − 4). State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

869

QUESTION ID
S24-M-Q4

TYPE
Short answer

CORRECT ANSWER
3; x ≠ −1, 4

GRADING NOTE Form required: result and all exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

Simplify $(5x^2 + 5x)/(x^2 - 4) \div (x + 1)/(x - 2)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

870

QUESTION ID
S24-M-Q5

TYPE
Short answer

CORRECT ANSWER
$5x/(x + 2)$; $x \neq -2, -1, 2$

GRADING NOTE Form required: result and all exclusions. $5x(x + 1)/((x - 2)(x + 2)) \cdot (x - 2)/(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 1

Multiply and divide fractions • Grade 6 • The Number System • 6.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2/3 · 9/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

872

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5/6 ÷ 10/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

873

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

True or false?

$a/b \div c/d = a/b \cdot c/d$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

874

QUESTION ID

S24-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Dividing multiplies by the reciprocal d/c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

3/8 ÷ 3/4 equals

A. 9/32

B. 2

C. 3/4

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

875

QUESTION ID

S24-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/2

GRADING NOTE 3/8 · 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 14/15 · 5/7.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

876

QUESTION ID

S24-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 2

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

878

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x − 2)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

Factor $5x^2 + 5x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

879

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5x(x + 1)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

880

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x − 2)(x + 2)

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$2x - 6 = 2(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

881

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Common factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

882

QUESTION ID

S24-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 3

Division by zero is undefined • Grade 7 • The Number System • 7.NS.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

For which x is 1/(x − 4) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

884

QUESTION ID

S24-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $(x + 2)/(x^2 - 9)$ undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

885

QUESTION ID

S24-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3 or −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

True or false?

After canceling (x − 3), x = 3 is allowed in the original expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

886

QUESTION ID

S24-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The original expression is undefined at x = 3, so it stays excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Dividing by (x + 1)/(x − 2) is impossible when

A. x = −1 only

B. x = 0

C. x = −1 or x = 2

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

887

QUESTION ID

S24-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = −1 or x = 2

GRADING NOTE Divisor undefined or equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

Why is $x = 0$ excluded from $(x + 2)/(3x^2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

888

QUESTION ID

S24-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The denominator $3x^2$ is 0 when $x = 0$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 1

Simplify complex fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (1/x + 1)/(1/x − 1). State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

890

QUESTION ID
S24-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(1 + x)/(1 − x); x ≠ 0, 1

GRADING NOTE Form required: result and exclusions. Multiply top and bottom by x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

Simplify $(1 - 4/x^2)/(1 + 2/x)$. State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

891

QUESTION ID
S24-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 2)/x;\ x \neq 0, -2$

GRADING NOTE Form required: result and exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

True or false?

$(a/b)/(c/d) = ad/(bc)$ when $b, c, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

892

QUESTION ID
S24-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Multiply by the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(x/2)/(x/4) for x ≠ 0 equals

A. 2

B. 1/2

C. $x^2/8$

D. $8/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

893

QUESTION ID

S24-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE x/2 · 4/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x - 1/x)/(x + 1)$. State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

894

QUESTION ID
S24-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$(x - 1)/x$; $x \neq 0, -1$

GRADING NOTE Form required: result and exclusions. $(x^2 - 1)/(x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 2

Explain closure of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the product of two rational expressions is a rational expression.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

896

QUESTION ID
S24-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(a/b)(c/d) = ac/(bd), and ac and bd are polynomials (bd not the zero polynomial).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

Explain why division by a nonzero rational expression gives a rational expression.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

897

QUESTION ID
S24-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(a/b) ÷ (c/d) = ad/(bc), a ratio of polynomials when c is not the zero polynomial.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

True or false?

Rational expressions are closed under division by the zero expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

898

QUESTION ID
S24-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

Which system is closed under addition, subtraction, multiplication, and division by nonzero elements?

A. Whole numbers

B. Rational expressions

C. Integers

D. Polynomials

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

899

QUESTION ID
S24-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Rational expressions

GRADING NOTE Like the rational numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

Give an example showing polynomials are not closed under division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

900

QUESTION ID
S24-F2-Q5

TYPE
Short answer

CORRECT ANSWER
x ÷ (x + 1) = x/(x + 1), which is not a polynomial

GRADING NOTE Any valid example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

Add and subtract rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Add fractions with unlike denominators	**5.NF.A.1**
BACKWARD 2	Find a least common denominator by factoring	**HSA-SSE.A.2**
BACKWARD 3	Subtract a whole numerator	**6.EE.A.3**
FORWARD 1	Decompose into partial fractions	**HSA-APR.D.7**
FORWARD 2	Solve rational equations	**HSA-REI.A.2**

SET 25 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x + 1)/(x^2 + x) + 1/x$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

903

QUESTION ID
S25-M-Q1

TYPE
Short answer

CORRECT ANSWER
$2/x$; $x \neq 0, -1$

GRADING NOTE Form required: result and all exclusions. $(x + 1)/(x(x + 1)) = 1/x$, but $x = -1$ stays excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2/(x + 1) + 3/(x − 2) as one rational expression.
State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

904

QUESTION ID
S25-M-Q2

TYPE
Short answer

CORRECT ANSWER
(5x − 1)/((x + 1)(x − 2)); x ≠ −1, 2

GRADING NOTE Form required: one fraction and exclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

Simplify x/(x − 3) − 3/(x − 3). State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

905

QUESTION ID
S25-M-Q3

TYPE
Short answer

CORRECT ANSWER
1; x ≠ 3

GRADING NOTE Form required: result and exclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q3**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

Write $1/(x^2 - 1) - 1/(x - 1)$ as one rational expression.
State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

906

QUESTION ID
S25-M-Q4

TYPE
Short answer

CORRECT ANSWER
$-x/((x - 1)(x + 1))$; $x \neq -1, 1$

GRADING NOTE Form required: one fraction and exclusions. $(1 - (x + 1))/((x - 1)(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

Write $4/(x^2 - 4x) - 1/x$ as one rational expression. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

907

QUESTION ID
S25-M-Q5

TYPE
Short answer

CORRECT ANSWER
$(8 - x)/(x(x - 4))$; $x \neq 0, 4$

GRADING NOTE Form required: one fraction and exclusions. $(4 - (x - 4))/(x(x - 4))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q5**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 1

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2/3 + 3/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

909

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

17/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

Compute 5/6 − 1/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

910

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

911

QUESTION ID

S25-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1/4 + 1/6 equals

A. 5/12

B. 2/10

C. 2/12

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

912

QUESTION ID

S25-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5/12

GRADING NOTE 3/12 + 2/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 3/10 + 2/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

913

QUESTION ID

S25-B1-Q5

TYPE

Short answer

CORRECT ANSWER

13/30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 2

Find a least common denominator by factoring • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x$ and $x^2 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

915

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x(x + 1); x(x - 4)$

GRADING NOTE Form required: both factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

What is the least common denominator of $1/(x^2 - 1)$ and $1/(x - 1)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

916

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

The LCD of $\frac{1}{x}$ and $\frac{1}{x(x-4)}$ is

A. $x - 4$

B. $x^2(x - 4)$

C. $x(x - 4)$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

917

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x(x - 4)$

GRADING NOTE Use each factor to its highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

True or false?

The LCD of $\frac{2}{x+1}$ and $\frac{3}{x-2}$ is $x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The denominators have no common factor; the LCD is $(x + 1)(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the LCD of $1/(x^2 + x)$ and $1/x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

919

QUESTION ID

S25-B2-Q5

TYPE

Short answer

CORRECT ANSWER

x(x + 1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 3

Subtract a whole numerator • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 4 − (x − 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

921

QUESTION ID

S25-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8 − x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify 1 − (x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

922

QUESTION ID

S25-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

$3 - (x - 3) = -x.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5

923

QUESTION ID

S25-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 - x + 3 = 6 - x.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5

$2(x - 2) - 3(x + 1)$ equals

A. $5x - 7$

B. $-x + 1$

C. $-x - 7$

D. $-x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

924

QUESTION ID

S25-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-x - 7$

GRADING NOTE $2x - 4 - 3x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $x - (3 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

925

QUESTION ID

S25-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2x − 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 1

Decompose into partial fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Find A and B so that
$(5x + 1)/((x - 1)(x + 2)) = A/(x - 1) + B/(x + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

927

QUESTION ID
S25-F1-Q1

TYPE
Short answer

CORRECT ANSWER
A = 2, B = 3

GRADING NOTE Form required: both. $5x + 1 = A(x + 2) + B(x - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find A and B so that 1/(x(x + 1)) = A/x + B/(x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

928

QUESTION ID
S25-F1-Q2

TYPE
Short answer

CORRECT ANSWER
A = 1, B = −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q2**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

True or false?

$1/(x(x + 1)) = 1/x + 1/(x + 1)$ for $x \neq 0, -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

929

QUESTION ID
S25-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1/x + 1/(x + 1) = (2x + 1)/(x(x + 1))$; the difference $1/x - 1/(x + 1)$ gives $1/(x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

$4/(x^2 - 4)$ equals

A. $2/(x - 2) - 2/(x + 2)$

B. $1/(x - 2) + 1/(x + 2)$

C. $4/(x - 2) - 4/(x + 2)$

D. $1/(x - 2) - 1/(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

930

QUESTION ID
S25-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D.
$1/(x - 2) - 1/(x + 2)$

GRADING NOTE Combine to check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $1/(n(n+1)) = 1/n - 1/(n+1)$ to find $1/(1\cdot2) + 1/(2\cdot3) + 1/(3\cdot4) + 1/(4\cdot5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

931

QUESTION ID
S25-F1-Q5

TYPE
Short answer

CORRECT ANSWER
4/5

GRADING NOTE Any equivalent form. Telescoping: $1 - 1/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 2

Solve rational equations • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2/(x + 1) + 3/(x − 2) = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

933

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/5

GRADING NOTE Any equivalent form. 5x − 1 = 0; not an excluded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

Solve x/(x − 3) − 3/(x − 3) = 1. Explain your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

934

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Every real x except 3 (the left side simplifies to 1 for x ≠ 3)

GRADING NOTE Form required: the solution set with the exclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

True or false?

A solution of a rational equation that makes a denominator 0 must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

935

QUESTION ID

S25-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solution of 1/x + 1/(2x) = 3 is

A. x = 1/2

B. x = 2

C. x = 3/2

D. no solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

936

QUESTION ID

S25-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x = 1/2

GRADING NOTE 3/(2x) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

Solve $4/(x^2 - 4x) - 1/x = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

937

QUESTION ID

S25-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. $(8 - x)/(x(x - 4)) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Create and solve rational, radical, and exponential equations from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Express rates of work and unit rates	**6.RP.A.3.b**
BACKWARD 2	Write exponential growth models	**HSF-LE.A.2**
BACKWARD 3	Solve linear equations after clearing a denominator	**HSA-REI.B.3**
FORWARD 1	Create equations whose algebraic solutions must be checked aga.	**HSA-CED.A.1**
FORWARD 2	Solve exponential context equations with logarithms	**HSF-LE.A.4**

SET 26 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

Ana can paint a room in 4 hours. Working together, Ana and Ben paint it in 2.4 hours. Write an equation for Ben's time b (in hours) working alone, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

940

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/4 + 1/b = 1/2.4;
b = 6 hours

GRADING NOTE Form required: the equation and the solution with units. 1/b = 5/12 − 3/12 = 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

A car's speed s (in mph) before braking is estimated from its skid length d (in feet) by $s = \sqrt{(22.5d)}$. Write and solve an equation for the skid length of a car going 45 mph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

941

QUESTION ID
S26-M-Q2

TYPE
Short answer

CORRECT ANSWER
$45 = \sqrt{(22.5d)}$;
d = 90 feet

GRADING NOTE Form required: the equation and the solution with units. 2025 = 22.5d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

A town of 1,200 people doubles in population every 5 years. Write an equation for the time t (in years) when the population is 9,600, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

942

QUESTION ID

S26-M-Q3

TYPE

Short answer

CORRECT ANSWER

$1200 \cdot 2^{t/5} = 9600$; t = 15 years

GRADING NOTE Form required: the equation and the solution with units. $2^{t/5} = 8 = 2^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A printer costs $500 plus $4 per shirt. Write an equation for the number of shirts x that makes the average cost per shirt $9, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

943

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

(500 + 4x)/x = 9;
x = 100 shirts

GRADING NOTE Form required: the equation and the solution. 500 = 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A tank holds 10 L of a 20% salt solution. Write an equation for the liters x of pure water to add to make an 8% solution, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

944

QUESTION ID
S26-M-Q5

TYPE
Short answer

CORRECT ANSWER
2/(10 + x) = 0.08;
x = 15 L

GRADING NOTE Form required: the equation and the solution with units. The salt stays 2 L; 10 + x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 1

Express rates of work and unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana paints a room in 4 hours. What fraction of the room does she paint per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

946

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

A pump fills a pool in 6 hours. What fraction does it fill in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

947

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

True or false?

If two people work together, their times to finish the job add.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

948

QUESTION ID

S26-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Their rates (jobs per hour) add; working together takes less time, not more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

A machine makes 300 parts in 5 hours. Its rate is

A. 1500 parts per hour

B. 300 parts per hour

C. 60 parts per hour

D. 5/300 parts per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

949

QUESTION ID

S26-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 60 parts per hour

GRADING NOTE 300/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

A job takes 2.4 hours together. What fraction is done per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

950

QUESTION ID

S26-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/2.4 = 5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 2

Write exponential growth models • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A population of 1,200 doubles every 5 years. Write the population after t years.

P(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

952

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$1200 \cdot 2^{t/5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A $500 deposit grows 3% per year. Write its value after t years.

V(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

953

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$500(1.03)^t$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

True or false?

A quantity that doubles every 5 years is multiplied by 2^5 after 20 years.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

954

QUESTION ID

S26-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20/5 = 4 doublings, a factor of $2^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

A quantity triples every year starting at 7. After t years it is

A. $7 + 3t$

B. $21t$

C. $3 \cdot 7^t$

D. $7 \cdot 3^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

955

QUESTION ID

S26-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $7 \cdot 3^t$

GRADING NOTE Repeated multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

How many doublings happen in 15 years if the doubling time is 5 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

956

QUESTION ID

S26-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 3

Solve linear equations after clearing a denominator • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 500 + 4x = 9x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

958

QUESTION ID

S26-B3-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2 = 0.08(10 + x).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

959

QUESTION ID

S26-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

True or false?

To solve (500 + 4x)/x = 9 we may multiply both sides by x when x ≠ 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

960

QUESTION ID

S26-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplying by a nonzero quantity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

2025 = 22.5d gives d =

A. 2002.5

B. 45

C. 9

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

961

QUESTION ID

S26-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 90

GRADING NOTE 2025/22.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5x/12 = 1/2.4 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

962

QUESTION ID

S26-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 1

Create equations whose algebraic solutions must be checked against the context • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A boat travels 24 km downstream and 24 km back in 5 hours. The current is 2 km/h. Write an equation for the boat's still-water speed b and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

964

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24/(b + 2) + 24/(b − 2) = 5; b = 10 km/h (reject b = −0.4)

GRADING NOTE Form required: equation, solution, and rejection. $5b^2 - 48b - 20 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The length of a rectangle is 3 more than its width w, and its diagonal is 15. Write an equation and find w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

965

QUESTION ID
S26-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$w^2 + (w + 3)^2 = 225$;
$w = 9$ (reject -12)

GRADING NOTE Form required: equation and solution with the rejection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

True or false?

A negative solution of an equation for a time must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

966

QUESTION ID
S26-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Time in the context is nonnegative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

A model gives t = −3 or t = 4 for the time (in seconds) of an event. The answer is

A. neither

B. both

C. 4 seconds

D. −3 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

967

QUESTION ID

S26-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4 seconds

GRADING NOTE Reject the negative time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe fills a tank in x hours and a second pipe takes 2 hours longer. Together they take 2.4 hours. Write an equation and find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

968

QUESTION ID

S26-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1/x + 1/(x + 2) = 1/2.4; x = 4 hours (reject −1.2)

GRADING NOTE Form required: equation and the solution. $2.4(2x + 2) = x(x + 2)$ gives $x^2 - 2.8x - 4.8 = 0$, so x = 4 or −1.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 2

Solve exponential context equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

The town of 1,200 doubles every 5 years. Write and solve an equation for when the population reaches 5,000. Give t to the nearest tenth of a year. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

970

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$1200 \cdot 2^{t/5} = 5000$; $t = 5 \log_2(25/6) \approx 10.3$ years

GRADING NOTE Form required: the equation and t (10.29 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

Solve $500(1.04)^t = 800$ for t. Give an exact answer and a decimal to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

971

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

t = log(1.6)/log(1.04) ≈ 11.98

GRADING NOTE Form required: exact form and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

True or false?

$2^{t/5} = 8$ can be solved without logarithms, but $2^{t/5} = 5$ cannot be solved exactly without them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

QUESTION ID

S26-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 8 is a power of 2; 5 is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

The exact solution of $3^x = 20$ is

A. 20/3

B. log(20/3)

C. log 20/log 3

D. 3 log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

973

QUESTION ID

S26-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. log 20/log 3

GRADING NOTE Take logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

Explain why log(1.6)/log(1.04) is the time for an amount growing 4% per year to grow by 60%.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

974

QUESTION ID

S26-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$(1.04)^t = 1.6$ is solved by t = log 1.6/log 1.04.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Create and solve nonlinear inequalities from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Sign of a product on intervals	**7.NS.A.2**
BACKWARD 3	Reverse the inequality when multiplying by a negative	**HSA-REI.B.3**
FORWARD 1	Solve rational inequalities with sign charts and excluded values	**HSA-CED.A.1**
FORWARD 2	Write systems of constraints and judge viability	**HSA-CED.A.3**

SET 27 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 64t$ feet after t seconds. Write and solve an inequality for the times when the ball is at least 48 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

977

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-16t^2 + 64t \geq 48$;
$1 \leq t \leq 3$ seconds

GRADING NOTE Form required: inequality and solution interval. $t^2 - 4t + 3 \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A company sells x items at a price of 100 − 2x dollars each. Write and solve an inequality for the values of x that give revenue of at least $1,200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

978

QUESTION ID
S27-M-Q2

TYPE
Short answer

CORRECT ANSWER
x(100 − 2x) ≥ 1200; 20 ≤ x ≤ 30

GRADING NOTE Form required: inequality and solution. (x − 20)(x − 30) ≤ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

The average cost of making x posters is (300 + 2x)/x dollars. Write and solve an inequality for the numbers of posters (x > 0) with an average cost of at most $5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

979

QUESTION ID
S27-M-Q3

TYPE
Short answer

CORRECT ANSWER
(300 + 2x)/x ≤ 5; x ≥ 100

GRADING NOTE Form required: inequality and solution. With x > 0, 300 + 2x ≤ 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Write and solve an inequality for the times when C is at least 4 mg/L.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

980

QUESTION ID
S27-M-Q4

TYPE
Short answer

CORRECT ANSWER
$20t/(t^2 + 4) \geq 4$;
$1 \leq t \leq 4$ hours

GRADING NOTE Form required: inequality and interval. $t^2 + 4 > 0$, so $20t \geq 4t^2 + 16$, $t^2 - 5t + 4 \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$500 is invested at 4% interest compounded yearly. Write an inequality for the number of whole years t until the value is at least $800, and solve it. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

981

QUESTION ID
S27-M-Q5

TYPE
Short answer

CORRECT ANSWER
$500(1.04)^t \geq 800$; $t \geq 11.98$, so 12 years

GRADING NOTE Form required: the inequality and the whole-year answer. Tables or logarithms both accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $t^2 - 4t + 3 = 0$.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

983

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 or 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 50x + 600 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

984

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20 or 30

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

True or false?

The solutions of $t^2 - 5t + 4 = 0$ are -1 and -4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

985

QUESTION ID
S27-B1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(t - 1)(t - 4) = 0$ gives $t = 1$ or $t = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

The solutions of $x^2 - x - 12 = 0$ are

A. 4 and −3

B. 6 and −2

C. 12 and −1

D. −4 and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

986

QUESTION ID

S27-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 and −3

GRADING NOTE $(x - 4)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2x^2 - 100x + 1200 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

987

QUESTION ID

S27-B1-Q5

TYPE

Short answer

CORRECT ANSWER

20 or 30

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 2

Sign of a product on intervals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is $(t - 1)(t - 3)$ positive or negative when $t = 2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

989

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Negative:

$(1)(-1) = -1$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $(t - 1)(t - 3)$ positive or negative when $t = 5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

990

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Positive: (4)(2) = 8

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

True or false?

A product of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

991

QUESTION ID

S27-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 20)(x - 30) \leq 0$ when

A. $x \leq 20$ or $x \geq 30$

B. $x \leq 20$

C. $20 \leq x \leq 30$

D. $x \geq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

992

QUESTION ID

S27-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $20 \leq x \leq 30$

GRADING NOTE Factors have opposite signs between the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which t is $(t - 1)(t - 4) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

993

QUESTION ID

S27-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$1 < t < 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 3

Reverse the inequality when multiplying by a negative • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-16t^2 + 64t - 48 \geq 0$ after dividing both sides by -16.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

995

QUESTION ID

S27-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$t^2 - 4t + 3 \leq 0$

GRADING NOTE Form required: the reversed inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-3x \geq 12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

996

QUESTION ID

S27-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x \leq -4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying both sides of an inequality by x is always allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

997

QUESTION ID
S27-B3-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only when the sign of x is known.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $x > 0$, then $(300 + 2x)/x \leq 5$ is equivalent to

A. $300 + 2x \leq 5x$

B. $300 + 2x \geq 5x$

C. $2x \leq 5$

D. $300 \leq 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

998

QUESTION ID

S27-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 300 + 2x ≤ 5x

GRADING NOTE Multiplying by positive x keeps the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2 - x < 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

999

QUESTION ID

S27-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x > -5$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 1

Solve rational inequalities with sign charts and excluded values • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 3)/(x + 1) \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

1001

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$-1 < x \leq 3$

GRADING NOTE Form required: the interval; −1 is excluded and 3 is included.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x + 2)/(x - 4) > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

1002

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x < -2$ or $x > 4$

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

True or false?

Multiplying $(x - 3)/(x + 1) \leq 0$ by $x + 1$ gives $x - 3 \leq 0$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

1003

QUESTION ID

S27-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The direction depends on the sign of $x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

The solution set of $1/x > 2$ is

A. $x > 1/2$

B. $x < 1/2$

C. $x > 0$

D. $0 < x < 1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

1004

QUESTION ID

S27-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $0 < x < 1/2$

GRADING NOTE For $x > 0$, $1 > 2x$; for $x < 0$ the left side is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model's average cost is (400 + 3x)/x dollars. For how many items x > 0 is the average cost under $7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

1005

QUESTION ID

S27-F1-Q5

TYPE

Short answer

CORRECT ANSWER

x > 100

GRADING NOTE Any equivalent form. 400 + 3x < 7x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 2

Write systems of constraints and judge viability • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

The ball's height must be at least 48 ft and the time at most 2.5 s. Write the constraints and find the viable times.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

1007

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-16t^2 + 64t \geq 48$ and $0 \leq t \leq 2.5$; viable times $1 \leq t \leq 2.5$

GRADING NOTE Form required: both constraints and the combined interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

Revenue x(100 − 2x) must be at least $1,200 and at most 24 items can be made. Which x are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

1008

QUESTION ID
S27-F2-Q2

TYPE
Short answer

CORRECT ANSWER
20 ≤ x ≤ 24 (whole numbers)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

True or false?

A value that solves the inequality may still not be viable in the context.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

1009

QUESTION ID
S27-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE For example a non-whole number of items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

Revenue $x(100 - 2x) \geq 1200$ with x a whole number is satisfied by

A. x = 19

B. x = 25

C. x = 31

D. x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

1010

QUESTION ID

S27-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = 25

GRADING NOTE $20 \leq x \leq 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

Explain why 12 years, not 11.98 years, answers the compound-interest question.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

1011

QUESTION ID
S27-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Interest is added once a year, so the value only reaches $800 at the end of a whole year.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

Create and graph nonlinear two-variable models

MAIN	Algebra II main question	**HSA-CED.A.2**
BACKWARD 1	Make a table of values from an equation	**6.EE.C.9**
BACKWARD 2	Choose scales and label axes	**HSN-Q.A.1**
BACKWARD 3	Write a growth or decay model from a percent	**HSF-LE.A.2**
FORWARD 1	Create equations in more than two variables and graph a member	**HSA-CED.A.2**
FORWARD 2	Interpret key features of the created models	**HSF-IF.B.4**

SET 28 • HSA-CED.A.2

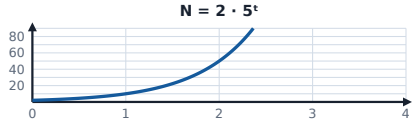
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

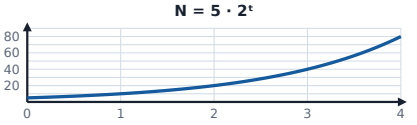
ANSWER KEY Reduced copy of the question page, with its answer

A rabbit population starts at 5 and doubles every year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 4$ (t in years, N rabbits)?

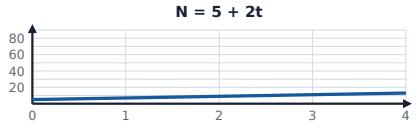
A



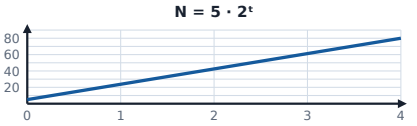
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

1014

QUESTION ID

S28-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Doubling every year from 5 gives $N = 5 \cdot 2^t$, an exponential curve through (0, 5), (1, 10), (2, 20), (3, 40), (4, 80). The others use the wrong model ($2 \cdot 5^t$ or $5 + 2t$) or pair the right equation with a straight line, which is not its graph.

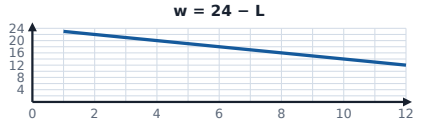
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

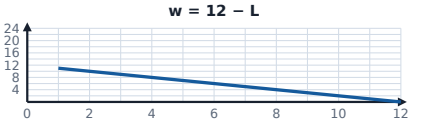
ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has area 24 square units. Which graph shows a correct equation for the situation with a correct graph of that equation for $1 \le L \le 12$ (length L, width w)?

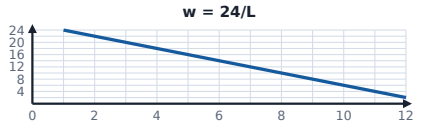
A



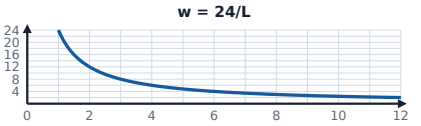
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

1015

QUESTION ID

S28-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

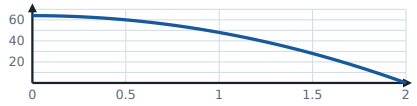
GRADING NOTE Area = Lw = 24 gives w = 24/L, a decreasing curve that flattens and approaches the L-axis, through (1, 24), (2, 12), (4, 6), (12, 2). The others use w = 24 − L, w = 12 − L (a perimeter idea), or draw a straight line for 24/L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

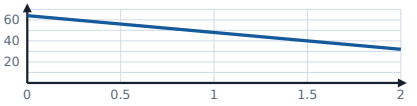
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

A stone is dropped from 64 feet. Its height after t seconds is 64 minus 16 times the square of t. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 2$ (h in feet)?

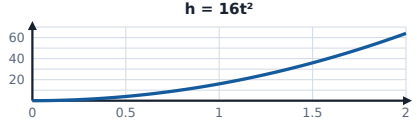
A



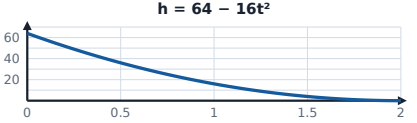
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

1016

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $h = 64 - 16t^2$: the height falls slowly at first and then faster, through (0, 64), (0.5, 60), (1, 48), (1.5, 28), (2, 0). The others use $64 - 16t$ (not squared), $16t^2$ (the distance fallen), or draw a curve that does not match $64 - 16t^2$ (it gives 16 at t = 1 instead of 48).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

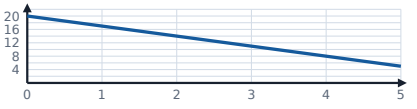
ANSWER KEY Reduced copy of the question page, with its answer

A $20,000 car loses 15% of its value each year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 5$ (V in thousands of dollars, t in years)?

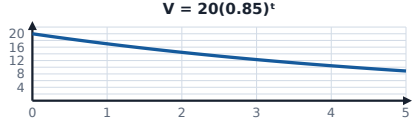
A



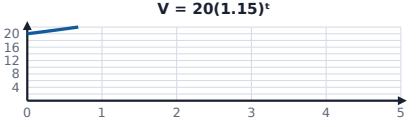
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

1017

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Keeping 85% each year gives $V = 20(0.85)^t$, through (1, 17), (2, 14.45), (5, 8.87). The others keep only 15% each year, subtract a fixed $3,000 each year (15% of the first value only), or grow 15% each year.

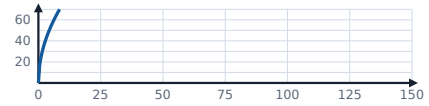
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

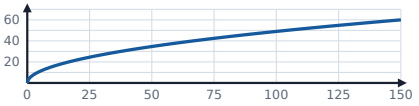
ANSWER KEY Reduced copy of the question page, with its answer

A car's speed s (mph) is the square root of 24 times its skid length d (ft). Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le d \le 150$?

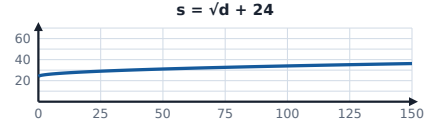
A



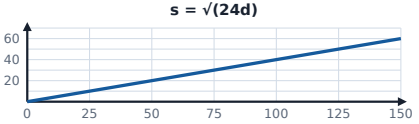
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

1018

QUESTION ID

S28-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE s = √(24d): an increasing curve that flattens, through (6, 12), (24, 24), (96, 48), (150, 60). The others use 24√d or √d + 24, or draw a straight line for √(24d) (it gives 24 at d = 60, not 37.9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 1

Make a table of values from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

For $N = 200 \cdot 2^t$, find N when t = 0, 1, 2, 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

1020

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

200, 400, 800, 1600

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For w = 24/L, find w when L = 2, 3, 4, 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

1021

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12, 8, 6, 4

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

True or false?

For $h = 64 - 16t^2$, $h = 48$ when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

1022

QUESTION ID

S28-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $h(2) = 64 - 64 = 0$; $h = 48$ when $t = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For s = √(24d), s when d = 96 is

A. 96

B. 48

C. 2304

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

1023

QUESTION ID

S28-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 48

GRADING NOTE √2304.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For $V = 20000(0.85)^t$, find V when t = 1 and t = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

1024

QUESTION ID

S28-B1-Q5

TYPE

Short answer

CORRECT ANSWER

17000 and 14450

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 2

Choose scales and label axes • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Values of N go from 200 to 3,200. Give a sensible step for the vertical axis on a grid with about 7 lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

1026

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

500

GRADING NOTE Any sensible step (for example 500).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quantity goes on the horizontal axis when V depends on t?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

1027

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

t (time in years)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Axis labels should include units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

1028

QUESTION ID

S28-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Units make the graph readable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The best vertical scale for values from 0 to 64 is

A. steps of 0.5

B. steps of 1 from 0 to 10

C. steps of 100 from 0 to 1000

D. steps of 10 from 0 to 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

1029

QUESTION ID
S28-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. steps of 10 from 0 to 70

GRADING NOTE Fits the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

1029

ANSWER KEY Reduced copy of the question page, with its answer

Give the horizontal range for $0 \le d \le 150$ with a step of 10. How many steps is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

1030

QUESTION ID

S28-B2-Q5

TYPE

Short answer

CORRECT ANSWER

15 steps

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 3

Write a growth or decay model from a percent • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the value of $20,000 after losing 15% per year for t years.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

1032

QUESTION ID

S28-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$20000(0.85)^t$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the growth factor for 4% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

1033

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1.04

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B3-Q2**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

True or false?

Losing 15% each year means multiplying by 0.15 each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

1034

QUESTION ID

S28-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply by $1 - 0.15 = 0.85$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

A quantity doubles every hour starting at 200. After t hours it is

A. $2 \cdot 200^t$

B. $200 \cdot 2^t$

C. 200 + 2t

D. 400t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

1035

QUESTION ID
S28-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $200 \cdot 2^t$

GRADING NOTE Doubling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the decay factor for 30% decay?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

1036

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B3-Q5**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 1

Create equations in more than two variables and graph a member of the family • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

The current I (amps) through a resistor is the voltage V divided by the resistance R. Write the equation, and write I as a function of R when V = 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

1038

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

I = V/R; I = 12/R

GRADING NOTE Form required: both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The volume of a cylinder is $V = \pi r^2 h$. Write h in terms of r when $V = 100\pi$, and describe its graph for $r > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

1039

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$h = 100/r^2$; decreasing, approaching 0 as r grows

GRADING NOTE Form required: the equation and a description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

True or false?

Fixing one variable of a three-variable equation gives a two-variable equation that can be graphed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

1040

QUESTION ID
S28-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A family of curves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

For $A = P(1.05)^t$ with $P = 1000$, the graph of A against t is

A. a line

B. an increasing exponential curve

C. a parabola

D. a decreasing curve

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

1041

QUESTION ID

S28-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. an increasing exponential curve

GRADING NOTE Growth factor 1.05 > 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

For the area formula $A = \pi r^2$, which graph (A against r) is correct for $r \geq 0$: a line, a half parabola, or an exponential?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

1042

QUESTION ID

S28-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Half of a parabola (from the origin, increasing)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 2

Interpret key features of the created models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

For $N = 200 \cdot 2^t$, what does the N-intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

1044

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

200 bacteria at the start (t = 0).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $h = 64 - 16t^2$, when does the stone hit the ground, and what is the t-intercept's meaning?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

1045

QUESTION ID
S28-F2-Q2

TYPE
Short answer

CORRECT ANSWER
t = 2 s; the time the height is 0

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

True or false?

The graph of $w = 24/L$ never reaches the L-axis because a rectangle of area 24 cannot have width 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

1046

QUESTION ID
S28-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $w > 0$ for every $L > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

For $V = 20000(0.85)^t$, the graph

A. decreases and approaches 0

B. increases

C. is a line

D. decreases to 0 at t = 6.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

1047

QUESTION ID

S28-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. decreases and approaches 0

GRADING NOTE Decay.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe the rate of change of $s = \sqrt{(24d)}$ as d grows.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

1048

QUESTION ID
S28-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It stays positive but decreases (the graph flattens).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

Represent nonlinear constraints and judge viable options

MAIN	Algebra II main question	**HSA-CED.A.3**
BACKWARD 1	Represent constraints with a linear system	**HSA-CED.A.3**
BACKWARD 2	Test whether a point satisfies an inequality	**6.EE.B.5**
BACKWARD 3	Solve quadratic inequalities	**HSA-CED.A.1**
FORWARD 1	Optimize within nonlinear constraints	**HSA-CED.A.3**
FORWARD 2	Locate constraint boundaries as graph intersections	**HSA-REI.D.11**

SET 29 • HSA-CED.A.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A garden has length L and width W (feet). The fencing for L + W is 30 ft and the area must be at least 200 ft². Write the constraints, find the viable widths, and decide whether W = 8 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

1051

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

L + W = 30 and LW ≥ 200; 10 ≤ W ≤ 20; W = 8 is not viable

GRADING NOTE Form required: constraints, the interval, and the decision. W(30 − W) ≥ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A box has a square base of side x and a volume of 32 in^3; both x and the height h must be at most 4 in. Write the constraints, find the viable x, and decide whether x = 3 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

1052

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2h = 32$, $x \le 4$, $h \le 4$; $2\sqrt{2} \le x \le 4$ (about 2.83 to 4); x = 3 is viable ($h \approx 3.56$)

GRADING NOTE Form required: constraints, the range, and the decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A plan invests $1,000 at 5% compounded yearly and needs at least $1,500 within 10 years. Write the constraint and decide whether the plan is viable. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

1053

QUESTION ID

S29-M-Q3

TYPE

Short answer

CORRECT ANSWER

$1000(1.05)^t \geq 1500$ for some $t \leq 10$; viable: $1000(1.05)^{10} \approx 1628.89$

GRADING NOTE Form required: the constraint and the decision with evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Attendance at a show is 500 − 10p people when tickets cost p dollars. Revenue must be at least $6,000. Write the constraints, find the viable prices, and decide whether $35 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

1054

QUESTION ID
S29-M-Q4

TYPE
Short answer

CORRECT ANSWER
p(500 − 10p) ≥ 6000, p ≥ 0; 20 ≤ p ≤ 30; $35 is not viable

GRADING NOTE Form required: constraints, interval, decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

A rectangular pen with sides x and y (m) needs area at least 50 m² and perimeter at most 40 m. Write the system and decide whether (5, 10) and (4, 12) are viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

1055

QUESTION ID
S29-M-Q5

TYPE
Short answer

CORRECT ANSWER
xy ≥ 50 and 2x + 2y ≤ 40; (5, 10) is viable (area 50, perimeter 30); (4, 12) is not (area 48)

GRADING NOTE Form required: the system and both decisions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 1

Represent constraints with a linear system • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write a constraint for "the total of L and W is 30".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

1057

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

L + W = 30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a constraint for "at most 40 m of fence for a rectangle with sides x and y".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

1058

QUESTION ID
S29-B1-Q2

TYPE
Short answer

CORRECT ANSWER
$2x + 2y \leq 40$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

"At least 200" is written ≤ 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

1059

QUESTION ID

S29-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At least 200 means 200 or more: ≥ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

1059

ANSWER KEY Reduced copy of the question page, with its answer

"The height h must be at most 4" is

A. $h \leq 4$

B. $h \geq 4$

C. $h < 4$

D. $h = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

1060

QUESTION ID

S29-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $h \leq 4$

GRADING NOTE At most: less than or equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (12, 20) a solution of x + y ≤ 30?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

1061

QUESTION ID

S29-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No: 32 > 30

GRADING NOTE Form required: the decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 2

Test whether a point satisfies an inequality • Grade 6 • Expressions & Equations • 6.EE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Does W = 8 satisfy W(30 − W) ≥ 200?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

1063

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

No: 8 · 22 = 176

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does p = 25 satisfy p(500 − 10p) ≥ 6000?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

1064

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 25 · 250 = 6250

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(4, 12) satisfies $xy \geq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

1065

QUESTION ID

S29-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $48 < 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which x satisfies $x^2 \geq 8$?

A. 3

B. 2

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

1066

QUESTION ID

S29-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $9 \geq 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Does t = 10 satisfy $1000(1.05)^t \geq 1500$? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

1067

QUESTION ID

S29-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: about 1628.89

GRADING NOTE Form required: decision and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 3

Solve quadratic inequalities • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve $W^2 - 30W + 200 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

1069

QUESTION ID

S29-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$10 \leq W \leq 20$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $p^2 - 50p + 600 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

1070

QUESTION ID

S29-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$20 \leq p \leq 30$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

True or false?

$x^2 \geq 8$ with $x > 0$ means $x \geq 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

1071

QUESTION ID

S29-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x \geq \sqrt{8} = 2\sqrt{2} \approx 2.83$; for example $x = 3$ works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

The solution of $(x - 2)(x - 6) \leq 0$ is

A. $x = 2$ or $x = 6$

B. $2 \leq x \leq 6$

C. $x \leq 2$ or $x \geq 6$

D. $x \geq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

1072

QUESTION ID

S29-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2 \leq x \leq 6$

GRADING NOTE Between the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $32/x^2 \leq 4$ for $x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

1073

QUESTION ID

S29-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 2\sqrt{2}$

GRADING NOTE Any equivalent form ($x \geq 2.83$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 1

Optimize within nonlinear constraints • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular garden has length L and width W (feet) with L + W = 30. Which width gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

1075

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

W = 15; 225 ft²

GRADING NOTE Form required: both. A = W(30 − W) has vertex at W = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

Which ticket price p gives the greatest revenue p(500 − 10p), and what is that revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

1076

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

p = $25; $6,250

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

True or false?

A maximum found from the model must still satisfy every other constraint to be a viable choice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

1077

QUESTION ID
S29-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Check all constraints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

For the pen with $xy \geq 50$ and $2x + 2y \leq 40$, which is viable?

A. (10, 10)

B. (15, 10)

C. (2, 20)

D. (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

1078

QUESTION ID

S29-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (10, 10)

GRADING NOTE (10, 10): area 100, perimeter 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

Among whole-number widths, list all viable garden widths when the area must be at least 216 ft².

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

1079

QUESTION ID

S29-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12, 13, 14, 15, 16, 17, 18

GRADING NOTE Form required: all. $W^2 - 30W + 216 \le 0$ gives $12 \le W \le 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 2

Locate constraint boundaries as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain how the viable widths $10 \le W \le 20$ can be read from the graphs of $y = W(30 - W)$ and $y = 200$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

1081

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The graphs intersect at W = 10 and W = 20; the parabola is above the line between them.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where do y = p(500 − 10p) and y = 6000 intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

1082

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

p = 20 and p = 30

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

True or false?

The x-coordinates where $y = 32/x^2$ meets $y = 4$ solve $32/x^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

1083

QUESTION ID

S29-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Intersection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 1000(1.05)^t$ and $y = 1500$ meet at $t \approx$

A. 15

B. 8.3

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

1084

QUESTION ID
S29-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 8.3

GRADING NOTE log 1.5/log 1.05 ≈ 8.31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

Explain why the region above y = 200 and below the parabola shows the viable widths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

1085

QUESTION ID

S29-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Those W give area W(30 − W) greater than or equal to 200.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

Rearrange rational, radical, and exponential formulas

MAIN	Algebra II main question	**HSA-CED.A.4**
BACKWARD 1	Rearrange linear formulas	**HSA-CED.A.4**
BACKWARD 2	Undo squares and square roots	**8.EE.A.2**
BACKWARD 3	Clear fractions in an equation	**HSA-REI.B.3**
FORWARD 1	Rearrange formulas where conditions decide the result	**HSA-CED.A.4**
FORWARD 2	Rearrange exponential formulas with logarithms	**HSF-LE.A.4**

SET 30 • HSA-CED.A.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

The lens formula is 1/f = 1/a + 1/b. Solve for b and state the condition needed.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

1088

QUESTION ID
S30-M-Q1

TYPE
Short answer

CORRECT ANSWER
af/(a − f), a ≠ f

GRADING NOTE Form required: the formula and the condition. 1/b = (a − f)/(af).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pendulum's period is $T = 2\pi\sqrt{(L/g)}$. Solve for L.

L = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

1089

QUESTION ID
S30-M-Q2

TYPE
Short answer

CORRECT ANSWER
$gT^2/(4\pi^2)$

GRADING NOTE Any equivalent form. Square both sides of $T/(2\pi) = \sqrt{(L/g)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An investment grows by $A = P(1 + r)^t$. Solve for r.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

1090

QUESTION ID
S30-M-Q3

TYPE
Short answer

CORRECT ANSWER
$(A/P)^{1/t} - 1$

GRADING NOTE Any equivalent form (the t-th root of A/P minus 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The speed of a falling object is $v = \sqrt{(2gh)}$. Solve for h and state the condition on v.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

1091

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

$v^2/(2g)$, with $v \geq 0$

GRADING NOTE Form required: formula and condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a circuit, I = V/(R + r). Solve for R and state the condition needed.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

1092

QUESTION ID
S30-M-Q5

TYPE
Short answer

CORRECT ANSWER
V/I − r, I ≠ 0

GRADING NOTE Form required: formula and condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 1

Rearrange linear formulas • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2L + 2W for W.

W = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

1094

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(P − 2L)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve d = rt for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

1095

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

d/r

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

True or false?

Solving V = IR for R gives R = VI.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

1096

QUESTION ID

S30-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide both sides by I: R = V/I.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

Solving y = mx + b for x gives

A. (y − b)/m

B. y/m − b

C. (y + b)/m

D. m(y − b)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

1097

QUESTION ID

S30-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (y − b)/m

GRADING NOTE Subtract b, divide by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve A = (1/2)bh for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

1098

QUESTION ID

S30-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2A/b

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 2

Undo squares and square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 49$ for $x > 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

1100

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{y} = 6$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

1101

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

True or false?

If √(L/g) = k, then L/g = √k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5

1102

QUESTION ID

S30-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Square both sides: L/g = k².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(\sqrt{(2gh)})^2$ equals

A. $2g\sqrt{h}$

B. $2gh$

C. $\sqrt{(2gh)}$

D. $4g^2h^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

1103

QUESTION ID

S30-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2gh

GRADING NOTE Squaring undoes the root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve T/(2π) = √m for m.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

1104

QUESTION ID

S30-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$T^2/(4\pi^2)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 3

Clear fractions in an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1/2 + 1/x = 1 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

1106

QUESTION ID

S30-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Combine 1/a − 1/f into one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

1107

QUESTION ID

S30-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(f − a)/(af)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $1/b = (a - f)/(af)$, then $b = af/(a - f)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

1108

QUESTION ID

S30-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving 3 = 12/(R + 2) for R gives

A. 36

B. 4

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

1109

QUESTION ID

S30-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE R + 2 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve I = V/(R + r) for R + r.

R + r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

1110

QUESTION ID

S30-B3-Q5

TYPE

Short answer

CORRECT ANSWER

V/I

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 1

Rearrange formulas where conditions decide the result • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = \pi r^2$ for r, given r is a length.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

1112

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(A/\pi)}$

GRADING NOTE Form required: the positive root only ($r \geq 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = (x + 1)/(x - 2)$ for x and state the condition on y.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

1113

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(2y + 1)/(y - 1)$, $y \neq 1$

GRADING NOTE Form required: formula and condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

True or false?

Solving $s = \sqrt{(24d)}$ for d gives $d = s^2/24$ for every real s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

1114

QUESTION ID
S30-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only $s \geq 0$ comes from the original formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

Solving $E = (1/2)mv^2$ for v ($v \geq 0$) gives

A. $\sqrt{(2E/m)}$

B. $(2E/m)^2$

C. $\sqrt{(E/2m)}$

D. $2E/m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

1115

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{(2E/m)}$

GRADING NOTE Multiply by 2/m and take the positive root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

Solve $P = k/V^2$ for V ($V > 0$).

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

1116

QUESTION ID

S30-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(k/P)}$

GRADING NOTE Form required: positive root with $k/P > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 2

Rearrange exponential formulas with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = Pe^{rt}$ for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

1118

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

ln(A/P)/r

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = P(1 + r)^t$ for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

1119

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

log(A/P)/log(1 + r)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $N = N_0 \cdot 2^{t/d}$ for t gives $t = d \cdot \log_2(N/N_0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

1120

QUESTION ID

S30-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take log base 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving $y = 5 \cdot 10^x$ for x gives

A. y/50

B. log(y/5)

C. $10^{y/5}$

D. log y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

1121

QUESTION ID

S30-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. log(y/5)

GRADING NOTE $10^x = y/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use t = log(A/P)/log(1 + r) to find t for A = 2000, P = 1000, r = 0.05. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

1122

QUESTION ID

S30-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14.21

GRADING NOTE Any value 14.2 to 14.21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSA-REI.A.2 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Solve rational equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Solve proportions	**7.RP.A.2**
BACKWARD 2	Find a common denominator of rational expressions	**HSA-APR.D.7**
BACKWARD 3	Values that make a denominator zero	**7.NS.A.2.b**
FORWARD 1	Explain and construct extraneous solutions	**HSA-REI.A.2**
FORWARD 2	See rational equations as graph intersections	**HSA-REI.D.11**

SET 31 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3/(x − 1) = 6/(x + 2). Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

1125

QUESTION ID
S31-M-Q1

TYPE
Short answer

CORRECT ANSWER
x = 4; none extraneous

GRADING NOTE Form required: solution and the check. 3x + 6 = 6x − 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

Solve x/(x − 2) + 1 = 2/(x − 2). Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

1126

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

No solution; x = 2 is extraneous

GRADING NOTE Form required: the conclusion and the extraneous value. 2x − 2 = 2 gives x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 1/x + 1/(x + 2) = 4/3. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

1127

QUESTION ID
S31-M-Q3

TYPE
Short answer

CORRECT ANSWER
x = 1 or x = −3/2; none extraneous

GRADING NOTE Form required: both solutions. $2x^2 + x - 3 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

Solve $x^2/(x - 3) = 9/(x - 3)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

1128

QUESTION ID

S31-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = −3; x = 3 is extraneous

GRADING NOTE Form required: solution and extraneous value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

Solve $2/(x + 1) - 1/(x - 1) = 2/(x^2 - 1)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

1129

QUESTION ID

S31-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = 5; none extraneous

GRADING NOTE Form required: solution and check. $2(x - 1) - (x + 1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 1

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3/x = 6/10.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

1131

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 4/7 = x/21.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

1132

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a/b = c/d$ implies $ac = bd$ when $b, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

1133

QUESTION ID

S31-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cross-multiplying gives ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

2/5 = 6/x gives x =

A. 12

B. 2.4

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

1134

QUESTION ID

S31-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

GRADING NOTE 2x = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3/(x - 1) = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

1135

QUESTION ID

S31-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 2

Find a common denominator of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

What is the LCD of 1/x and 1/(x + 2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

1137

QUESTION ID
S31-B2-Q1

TYPE
Short answer

CORRECT ANSWER
x(x + 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

What is the LCD of $\frac{2}{(x + 1)}$, $\frac{1}{(x - 1)}$, and $\frac{2}{(x^2 - 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

1138

QUESTION ID
S31-B2-Q2

TYPE
Short answer

CORRECT ANSWER
(x + 1)(x − 1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

True or false?

The LCD of $1/(x-2)$ and $2/(x-2)$ is $(x-2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

1139

QUESTION ID
S31-B2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is $x-2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiplying $1/x + 1/(x + 2) = 4/3$ by $3x(x + 2)$ gives

A. $3 + 3 = 4$

B. $3(x + 2) + 3x = 4$

C. $3(x + 2) + 3x = 4x(x + 2)$

D. $x + 2 + x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

1140

QUESTION ID
S31-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C.
$3(x + 2) + 3x = 4x(x + 2)$

GRADING NOTE Each term times the LCD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/x + 1/(x + 2) as one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

1141

QUESTION ID
S31-B2-Q5

TYPE
Short answer

CORRECT ANSWER
(2x + 2)/(x(x + 2))

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 3

Values that make a denominator zero • Grade 7 • The Number System • 7.NS.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

Which x-values make x/(x − 2) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

1143

QUESTION ID

S31-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which x-values make $2/(x^2 - 1)$ undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

1144

QUESTION ID

S31-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1 or −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

True or false?

A rational equation cannot have a solution that makes any denominator zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

1145

QUESTION ID

S31-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The expression is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

Which value is excluded from 1/x + 1/(x + 2) = 4/3?

A. 2

B. 1

C. 4/3

D. −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

1146

QUESTION ID

S31-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −2

GRADING NOTE x + 2 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the excluded values of $x^2/(x - 3) = 9/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

1147

QUESTION ID

S31-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 1

Explain and construct extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Explain how multiplying both sides of $x/(x-2) + 1 = 2/(x-2)$ by $x-2$ creates an extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

1149

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Multiplying by $x-2$ is not reversible when $x-2=0$, so the new equation is true at $x=2$ although the original is undefined there.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

Write a rational equation that has the extraneous solution x = 4 and no actual solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

1150

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

For example

x/(x − 4) = 4/(x − 4)

GRADING NOTE Any valid example with a check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

True or false?

Every rational equation has at least one extraneous solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

1151

QUESTION ID

S31-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3/(x − 1) = 6/(x + 2) has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

For which a does a/(x − 1) = 2/(x − 1) have infinitely many solutions?

A. a = 2

B. no value

C. a = 0

D. a = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

1152

QUESTION ID

S31-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. a = 2

GRADING NOTE Then every x ≠ 1 works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve (x + 1)/(x − 1) = 2/(x − 1) + 1 and explain the result.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

1153

QUESTION ID
S31-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Every x ≠ 1 is a solution (the equation is an identity on its domain).

GRADING NOTE Any equivalent answer. (x + 1)/(x − 1) − 2/(x − 1) = (x − 1)/(x − 1) = 1 for x ≠ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 2

See rational equations as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain why $x/(x - 2) + 1 = 2/(x - 2)$ has no solution using the graphs of $y = x/(x - 2) + 1$ and $y = 2/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

1155

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

For $x \neq 2$ the left side minus the right side is $(2x - 4)/(x - 2) = 2$, so the graphs never meet; at $x = 2$ neither is defined.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

Where do y = 3/(x − 1) and y = 6/(x + 2) intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

1156

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 1)

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

True or false?

An extraneous solution corresponds to an intersection point of the two graphs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

1157

QUESTION ID

S31-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Neither graph is defined there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = x^2/(x - 3)$ and $y = 9/(x - 3)$ intersect at

A. (3, 3/2)

B. (−3, 3/2)

C. (−3, −3/2)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

1158

QUESTION ID

S31-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, −3/2)

GRADING NOTE $9/(-6) = -3/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to approximate the solution of $1/x = x - 1$ with $x > 0$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

1159

QUESTION ID

S31-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1.62

GRADING NOTE Any value 1.61 to 1.62 ($(1 + \sqrt{5})/2 \approx 1.618$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSA-REI.A.2 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Solve radical equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Square and cube both sides; principal roots	**8.EE.A.2**
BACKWARD 2	Solve the resulting quadratic	**HSA-REI.B.4.b**
BACKWARD 3	Check a candidate in the original equation	**6.EE.B.5**
FORWARD 1	Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$	**HSA-REI.A.2**
FORWARD 2	Explain extraneous solutions with graphs of radical functions	**HSF-IF.C.7.b**

SET 32 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 7) = x − 5. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

1162

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9; x = 2 is extraneous

GRADING NOTE Form required: solution and extraneous value. $x^2 - 11x + 18 = 0$; $\sqrt{9} = 3 \neq -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(2x − 1) = 3. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

1163

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 5; none extraneous

GRADING NOTE Form required: solution and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

Solve √(x + 3) + 3 = x. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

1164

QUESTION ID

S32-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = 6; x = 1 is extraneous

GRADING NOTE Form required: solution and extraneous value. $x^2 - 7x + 6 = 0$; √4 + 3 = 5 ≠ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

Solve $\sqrt[3]{(x - 2)} = -3$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

1165

QUESTION ID

S32-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = −25; none extraneous

GRADING NOTE Form required: solution. Cubing is reversible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q4**

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

Solve √(x − 1) + 5 = 2. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

1166

QUESTION ID

S32-M-Q5

TYPE

Short answer

CORRECT ANSWER

No solution; x = 10 is extraneous

GRADING NOTE Form required: conclusion. √(x − 1) = −3 is impossible; squaring gives x = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 1

Square and cube both sides; principal roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{y} = 3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

1168

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt[3]{y} = -3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

1169

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−27

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

True or false?

$\sqrt{9} = 3$ or -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5

1170

QUESTION ID

S32-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The principal square root is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{y} = -2$ has

A. $y = 2$

B. no solution

C. $y = 4$

D. $y = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

1171

QUESTION ID

S32-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. no solution

GRADING NOTE A principal square root is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Square both sides of $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

1172

QUESTION ID

S32-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x + 7 = x^2 - 10x + 25$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 2

Solve the resulting quadratic • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 11x + 18 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

1174

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 or 9

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 7x + 6 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

1175

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 or 6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)^2 = x^2 + 25.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

1176

QUESTION ID

S32-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 10x + 25.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

$x^2 - 6x + 9 = x + 3$ gives

A. $x^2 - 7x + 12 = 0$

B. $x^2 - 5x + 12 = 0$

C. $x^2 - 6x + 6 = 0$

D. $x^2 - 7x + 6 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

1177

QUESTION ID

S32-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 7x + 6 = 0$

GRADING NOTE Subtract x + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 4x = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

1178

QUESTION ID

S32-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0 or 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 3

Check a candidate in the original equation • Grade 6 • Expressions & Equations • 6.EE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 2$ in $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

1180

QUESTION ID

S32-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{9} = 3$ but $2 - 5 = -3$; not a solution

GRADING NOTE Form required: the check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Check x = 6 in √(x + 3) + 3 = x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

1181

QUESTION ID

S32-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√9 + 3 = 6; a solution

GRADING NOTE Form required: the check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

True or false?

$x = 4$ satisfies $\sqrt{(2x - 1)} = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

1182

QUESTION ID

S32-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(8 - 1)} = \sqrt{7} \neq 3$; the solution is $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value satisfies √(x + 3) + 3 = x?

A. 1

B. 0

C. 6

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

1183

QUESTION ID

S32-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Check each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 10$ in $\sqrt{(x - 1)} + 5 = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

1184

QUESTION ID

S32-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3 + 5 = 8 ≠ 2; not a solution

GRADING NOTE Form required: check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 1

Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$ • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{(x + 5)} - \sqrt{x} = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

1186

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $\sqrt{(x + 5)} = 1 + \sqrt{x}$ gives $\sqrt{x} = 2$; check $\sqrt{9} - 2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^{3/2} = 8$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

1187

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $x = 8^{2/3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{(x^2)} = 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

1188

QUESTION ID

S32-F1-Q3

TYPE

Short answer

CORRECT ANSWER

5 or −5

GRADING NOTE Form required: both. $\sqrt{(x^2)} = |x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

1189

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$; for example $x = -3$ gives 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

The solution set of $\sqrt{(x^2)} = x$ is

A. $x \geq 0$

B. $x \leq 0$

C. $x = 0$ only

D. all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

1190

QUESTION ID

S32-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. x ≥ 0

GRADING NOTE $|x| = x$ exactly when $x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 2

Explain extraneous solutions with graphs of radical functions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = \sqrt{(x + 7)}$ and $y = x - 5$ meet at one point. Give it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

1192

QUESTION ID
S32-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(9, 4)

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The extraneous value $x = 2$ is where $y = x - 5$ meets which other graph?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

1193

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y = -\sqrt{(x + 7)}$

GRADING NOTE Any equivalent answer. Squaring cannot tell √ from −√.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

True or false?

The graph of $y = \sqrt{(x + 7)}$ starts at (7, 0) and increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

1194

QUESTION ID
S32-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x + 7 = 0$ at $x = -7$, so it starts at $(-7, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = \sqrt{(x - 1)} + 5$ lies

A. below $y = 2$

B. below the x-axis

C. at or below $y = 5$

D. at or above $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

1195

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. at or above y = 5

GRADING NOTE $\sqrt{} \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the graphs to explain why $\sqrt{(x - 1)} + 5 = 2$ has no solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

1196

QUESTION ID

S32-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$y = \sqrt{(x - 1)} + 5$ is always at least 5, so it never meets $y = 2$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

Explain why intersections of nonlinear graphs solve f(x) = g(x)

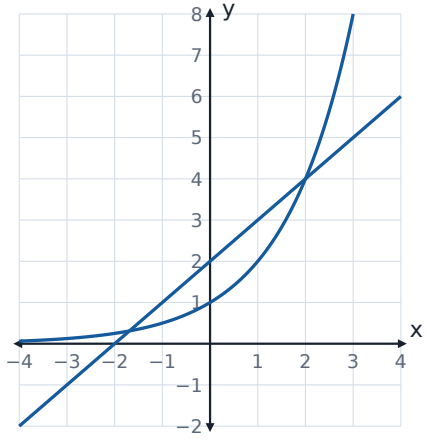
MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	A graph is the set of its solutions	**HSA-REI.D.10**
BACKWARD 2	Solve a linear system by graphing	**8.EE.C.8.a**
BACKWARD 3	Evaluate nonlinear functions at an input	**HSF-IF.A.2**
FORWARD 1	Relate intersections to zeros of f - g	**HSA-REI.D.11**
FORWARD 2	Find all solutions of a trigonometric equation on an interval from a…	**HSF-TF.B.7**

SET 33 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $y = 2^x$ and $y = x + 2$ are shown. They meet at $x = 2$ and at a negative x. Explain why $x = 2$ solves $2^x = x + 2$, and verify it.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

1199

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

At an intersection both functions have the same y-value for the same x, so $2^x = x + 2$ there; $2^2 = 4 = 2 + 2$.

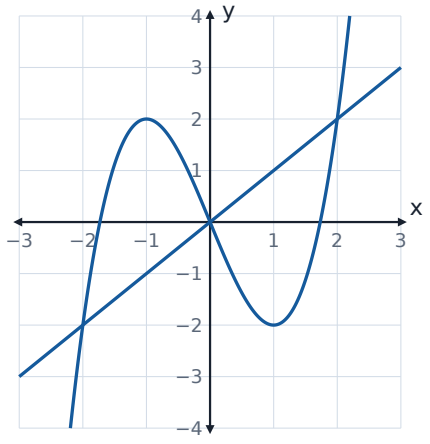
GRADING NOTE Form required: the explanation and the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

The graphs of $y = x^3 - 3x$ and $y = x$ are shown. Use them to list every solution of $x^3 - 3x = x$, and explain why they are solutions.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

1200

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = −2, 0, 2: at these x-values the two outputs are equal (each point lies on both graphs).

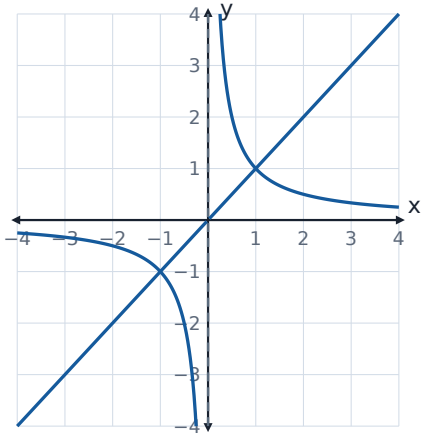
GRADING NOTE Form required: all three solutions and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 1/x$ and $y = x$ are shown. List every solution of $1/x = x$ and explain using the graphs.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

1201

QUESTION ID
S33-M-Q3

TYPE
Short answer

CORRECT ANSWER
x = −1 and x = 1: the graphs meet at (−1, −1) and (1, 1), where 1/x and x are equal.

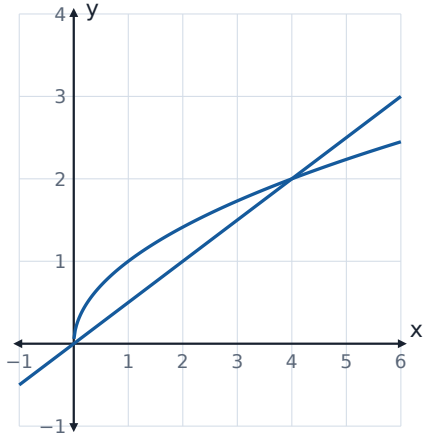
GRADING NOTE Form required: both solutions and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

The graphs of $y = \sqrt{x}$ and $y = x/2$ are shown. List every solution of $\sqrt{x} = x/2$ and verify each.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

1202

QUESTION ID

S33-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = 0$ and $x = 4$: $\sqrt{0} = 0 = 0/2$ and $\sqrt{4} = 2 = 4/2$.

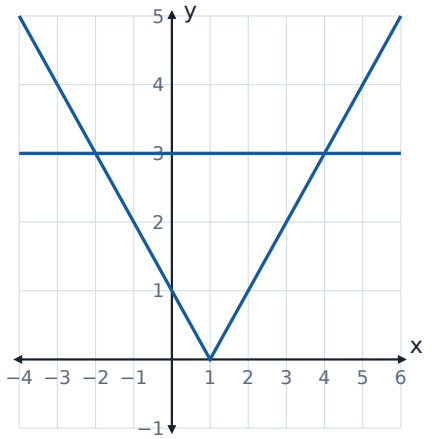
GRADING NOTE Form required: both solutions and the checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

The graphs of $y = |x - 1|$ and $y = 3$ are shown. List every solution of $|x - 1| = 3$ and explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

1203

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = −2 and x = 4: the V-shaped graph is at height 3 there.

GRADING NOTE Form required: both solutions and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 1

A graph is the set of its solutions • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 4) on the graph of $y = 2^x$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

1205

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: $2^2 = 4$

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (3, 6) on the graph of y = x + 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

1206

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No: 3 + 2 = 5

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

True or false?

Every point on the graph of y = f(x) has the form (x, f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

1207

QUESTION ID
S33-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition of the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on $y = x^3 - 3x$?

A. (0, 3)

B. (2, 6)

C. (2, 2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

1208

QUESTION ID

S33-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 2)

GRADING NOTE $8 - 6 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (−1, −1) on the graph of y = 1/x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

1209

QUESTION ID

S33-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: 1/(−1) = −1

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 2

Solve a linear system by graphing • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

The lines y = x + 1 and y = −x + 5 meet at (2, 3).
Explain why x = 2 solves x + 1 = −x + 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

1211

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

At x = 2 both equal 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where do y = 2x and y = 6 meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

1212

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 6)

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

True or false?

Two lines that do not intersect give an equation with no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

1213

QUESTION ID

S33-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 3$ and $y = x - 1$ meet at $x =$

A. 4

B. 3

C. -1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

1214

QUESTION ID

S33-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $x - 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve x + 1 = −x + 5 algebraically.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

1215

QUESTION ID

S33-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 3

Evaluate nonlinear functions at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find 2^x and $x + 2$ when $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

1217

QUESTION ID

S33-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 and 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $x^3 - 3x$ when $x = -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

1218

QUESTION ID

S33-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{9} = 9/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

1219

QUESTION ID

S33-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{9} = 3$, but $9/2 = 4.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$|x - 1|$ when $x = -2$ is

A. 3

B. 1

C. -1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

1220

QUESTION ID

S33-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $|-3|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 1/x when x = −1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

1221

QUESTION ID

S33-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 1

Relate intersections to zeros of f − g • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the solutions of $x^3 - 3x = x$ are the zeros of $h(x) = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

1223

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(x) = g(x) exactly when f(x) − g(x) = 0, and f − g = x³ − 4x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

Find the zeros of $h(x) = x^3 - 4x$ algebraically.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

1224

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−2, 0, 2

GRADING NOTE Form required: all three. x(x − 2)(x + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

True or false?

If y = f(x) and y = g(x) meet at x = a, then the graph of y = f(x) − g(x) crosses or touches the x-axis at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

1225

QUESTION ID

S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(a) − g(a) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solutions of $\sqrt{x} = x/2$ are the zeros of

A. $\sqrt{x} + x/2$

B. $\sqrt{x} - x/2$

C. $\sqrt{x} \cdot x/2$

D. $x/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q4/5

1226

QUESTION ID
S33-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $\sqrt{x} - x/2$

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q4/5

Explain why $2^x = x + 2$ has exactly two solutions using $h(x) = 2^x - x - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

1227

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

h changes sign twice (h(−2) > 0, h(0) < 0, h(3) > 0) and h is decreasing then increasing, so it has exactly two zeros.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 2

Find all solutions of a trigonometric equation on an interval from a graph • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

On $0 \le x < 2\pi$, where does $y = \sin x$ meet $y = 1/2$? Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

1229

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On $0 \le x < 2\pi$, where does $y = \cos x$ meet $y = 0$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

1230

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

π/2 and 3π/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

sin x = 1/2 has exactly two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

1231

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has two solutions in each period, so infinitely many real solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

How many solutions does $\sin x = 0.3$ have on $0 \le x < 2\pi$?

A. 1

B. 0

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

1232

QUESTION ID

S33-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE The horizontal line crosses twice in one period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why sin x = 2 has no solution using the graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

1233

GRADING NOTE Any equivalent explanation.

QUESTION ID

S33-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The graph of y = sin x stays between −1 and 1, so it never meets y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

Approximate solutions of f(x) = g(x)

MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	Read solutions of linear-quadratic systems from graphs	**HSA-REI.D.11**
BACKWARD 2	Evaluate exponential and logarithmic expressions with a calculat…	**HSF-IF.A.2**
BACKWARD 3	Round to a place value	**5.NBT.A.4**
FORWARD 1	Guarantee the accuracy of an approximation	**HSA-REI.D.11**
FORWARD 2	Compare an approximation with an exact logarithmic solution	**HSF-LE.A.4**

SET 34 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Use a table or successive approximation to solve $2^x = 5 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

1236

QUESTION ID
S34-M-Q1

TYPE
Short answer

CORRECT ANSWER
1.72

GRADING NOTE Accept 1.71 to 1.72 (exact root 1.7156...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 7 - x$ to the nearest hundredth with a table or successive approximation. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

1237

QUESTION ID
S34-M-Q2

TYPE
Short answer

CORRECT ANSWER
1.74

GRADING NOTE Accept 1.73 to 1.74 (root 1.7392...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x = 2 − x (log base 10) to the nearest hundredth. A calculator may be used.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

1238

QUESTION ID
S34-M-Q3

TYPE
Short answer

CORRECT ANSWER
1.76

GRADING NOTE Accept 1.75 to 1.76 (root 1.7556...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows $f(x) = 3(0.8)^x$ and $g(x) = x$. Between which two table values is the solution of $f(x) = g(x)$? Estimate it to the nearest tenth.

x	0	0.5	1	1.5	2	2.5
f(x)	3	2.683	2.4	2.147	1.92	1.717
g(x)	0	0.5	1	1.5	2	2.5

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

1239

QUESTION ID
S34-M-Q4

TYPE
Short answer

CORRECT ANSWER
Between 1.5 and 2; x ≈ 1.9

GRADING NOTE Form required: the interval and the estimate (root 1.944...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to find the negative solution of $2^x = x + 2$ to the nearest tenth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

1240

QUESTION ID
S34-M-Q5

TYPE
Short answer

CORRECT ANSWER
−1.7

GRADING NOTE Any equivalent form (root −1.690...).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 1

Read solutions of linear-quadratic systems from graphs • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Where do $y = x^2$ and $y = 4$ meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

1242

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x = −2 and x = 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Read the intersection of $y = x^2$ and $y = x + 2$ from a graph or table.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

1243

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 and x = 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The solutions of f(x) = g(x) can be approximated from a table where f − g changes sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

1244

QUESTION ID

S34-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sign change between rows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) - g(1) = -2$ and $f(2) - g(2) = 1$. A solution lies

A. below 1

B. at 1.5 exactly

C. above 2

D. between 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

1245

QUESTION ID

S34-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. between 1 and 2

GRADING NOTE Sign change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = 3$. Between which consecutive integers is the positive solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

1246

QUESTION ID
S34-B1-Q5

TYPE
Short answer

CORRECT ANSWER
1 and 2

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 2

Evaluate exponential and logarithmic expressions with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find $2^{1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

1248

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find log 1.75 (base 10) to the nearest thousandth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

1249

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.243

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1.7^3 = 5.1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

1250

QUESTION ID

S34-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1.7 \cdot 1.7 \cdot 1.7 = 4.913$ (5.1 is $3 \cdot 1.7$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3(0.8)^2$ equals

A. 0.192

B. 4.8

C. 1.92

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

1251

QUESTION ID

S34-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1.92

GRADING NOTE $3 \cdot 0.64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $2^{-1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

1252

QUESTION ID

S34-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.31

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 3

Round to a place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Round 1.7156 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

1254

QUESTION ID

S34-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.72

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 1.944 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

1255

QUESTION ID

S34-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1.9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

1.755 rounded to the nearest hundredth is 1.76.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

1256

QUESTION ID

S34-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Round half up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

-1.690 to the nearest tenth is

A. -1.6

B. -1.69

C. -2

D. -1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

1257

QUESTION ID

S34-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. -1.7

GRADING NOTE Nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 1.7392 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

1258

QUESTION ID

S34-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1.74

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 1

Guarantee the accuracy of an approximation • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

$h(x) = 2^x + x - 5$. Find h(1.71) and h(1.72) and explain why a solution lies between them. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

1260

QUESTION ID
S34-F1-Q1

TYPE
Short answer

CORRECT ANSWER
h(1.71) ≈ −0.018 and h(1.72) ≈ 0.014; h changes sign, so h has a zero between them

GRADING NOTE Form required: both values and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

Use one step of bisection on [1, 2] for $x^3 + x - 7 = 0$. Which half contains the solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

1261

QUESTION ID
S34-F1-Q2

TYPE
Short answer

CORRECT ANSWER
[1.5, 2], since h(1.5) = −2.125 < 0 and h(2) = 3 > 0

GRADING NOTE Form required: the half and the evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

True or false?

After 4 bisection steps starting from an interval of length 1, the interval has length 1/8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

1262

QUESTION ID

S34-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each step halves the length: $(1/2)^4 = 1/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

How many bisection steps from [1, 2] guarantee an interval shorter than 0.01?

A. 5

B. 100

C. 10

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

1263

QUESTION ID

S34-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE $2^7 = 128 > 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why checking h(1.715) and h(1.725) decides whether the solution rounds to 1.72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

1264

QUESTION ID

S34-F1-Q5

TYPE

Short answer

CORRECT ANSWER

If h changes sign between 1.715 and 1.725, the zero is in that interval, and every number there rounds to 1.72.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 2

Compare an approximation with an exact logarithmic solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 5$ exactly, then give a decimal to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

1266

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2 5 = \log 5/\log 2 \approx 2.32$

GRADING NOTE Form required: exact and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3(0.8)^x = 1$ exactly and to the nearest hundredth.
A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

1267

QUESTION ID
S34-F2-Q2

TYPE
Short answer

CORRECT ANSWER
log(1/3)/log(0.8) ≈ 4.92

GRADING NOTE Form required: exact and decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

True or false?

$2^x = 5 - x$ can be solved exactly with one logarithm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

1268

QUESTION ID

S34-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x appears both in an exponent and outside it; approximation is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation can be solved exactly with logarithms?

A. $\log x = 2 - x$

B. $2^x = x + 2$

C. $5(1.2)^x = 30$

D. $3(0.8)^x = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

1269

QUESTION ID

S34-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $5(1.2)^x = 30$

GRADING NOTE Only one exponential term and no other x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

Explain why approximation methods are needed for $2^x = x + 2$ but not for $2^x = 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

1270

QUESTION ID

S34-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2^x = 4$ isolates the exponential, so $x = \log_2 4 = 2$ exactly; $2^x = x + 2$ mixes an exponential and a linear term, which no algebra step separates.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSF-TF.A.1 • TRIGONOMETRIC FUNCTIONS

Use radian measure as arc length on the unit circle

MAIN	Algebra II main question	**HSF-TF.A.1**
BACKWARD 1	Circumference of a circle	**7.G.B.4**
BACKWARD 2	Arc length is proportional to the angle	**HSG-C.B.5**
BACKWARD 3	Degrees as fractions of a turn	**4.MD.C.5**
FORWARD 1	Measure negative angles and angles beyond one turn	**HSF-TF.A.1**
FORWARD 2	Read sine and cosine of radian angles from the unit circle	**HSF-TF.A.2**

SET 35 • HSF-TF.A.1

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

On the unit circle, a central angle subtends an arc of length 2.5. What is the radian measure of the angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

1273

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.5 radians: on the unit circle the radian measure equals the length of the subtended arc.

GRADING NOTE Form required: the measure and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 150° to radians. Give an exact answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

1274

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

5π/6

GRADING NOTE Form required: exact. 150 · π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 7π/4 radians to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

1275

QUESTION ID

S35-M-Q3

TYPE

Short answer

CORRECT ANSWER

315°

GRADING NOTE Any equivalent form. (7/4)·180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle of 2π/3 radians is drawn in a circle of radius 6 cm. Find the length of the arc it subtends.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

1276

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

4π cm (about 12.57 cm)

GRADING NOTE Form required: exact or decimal with units. s = rθ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why one full turn measures 2π radians.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

1277

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

A full turn subtends the whole unit circle, whose circumference is 2π · 1 = 2π.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 1

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle of radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

1279

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle of radius 6 cm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

1280

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12π cm

GRADING NOTE Any equivalent form with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

True or false?

The circumference of a circle is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

1281

QUESTION ID

S35-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE πr^2 is the area; the circumference is $\pi d = 2\pi r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

Half of the unit circle has length

A. 2π

B. 1

C. π

D. π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

1282

QUESTION ID

S35-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π

GRADING NOTE 2π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What length is one quarter of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

1283

QUESTION ID

S35-B1-Q5

TYPE

Short answer

CORRECT ANSWER

π/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 2

Arc length is proportional to the angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A 60° angle subtends what fraction of a full circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

1285

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q1**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the arc length of a 90° angle in a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

1286

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

True or false?

Doubling the radius doubles the arc length for the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

1287

QUESTION ID

S35-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Arc length is proportional to the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

The arc length for angle θ (radians) in a circle of radius r is

A. θ/r

B. $r^2\theta$

C. $2\pi\theta$

D. rθ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

1288

QUESTION ID

S35-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. rθ

GRADING NOTE Definition of radian measure scaled by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of the circle is an angle of π/3 radians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

1289

QUESTION ID

S35-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form. (π/3)/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q5**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 3

Degrees as fractions of a turn • Grade 4 • Measurement & Data • 4.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

How many degrees are in a full turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

1291

QUESTION ID

S35-B3-Q1

TYPE

Short answer

CORRECT ANSWER

360°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a turn is 150°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

1292

QUESTION ID

S35-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

True or false?

A 315° angle is 3/4 of a turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

1293

QUESTION ID

S35-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 315/360 = 7/8; 3/4 of a turn is 270°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A quarter turn is

A. 90°

B. 180°

C. 45°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

1294

QUESTION ID

S35-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

GRADING NOTE 360/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

How many degrees are in 2/3 of a turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

1295

QUESTION ID

S35-B3-Q5

TYPE

Short answer

CORRECT ANSWER

240°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 1

Measure negative angles and angles beyond one turn • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Give a positive angle less than 2π that is coterminal with $-\pi/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

1297

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$5\pi/3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the angle between 0 and 2π that is coterminal with 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

1298

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

True or false?

An arc of length 10 on the unit circle goes more than once around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

1299

QUESTION ID

S35-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10 > 2\pi \approx 6.28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle of $-\pi/2$ radians is traced

A. a quarter turn clockwise

B. one full turn

C. half a turn

D. a quarter turn counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

1300

QUESTION ID

S35-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. a quarter turn clockwise

GRADING NOTE Negative means clockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

1300

ANSWER KEY Reduced copy of the question page, with its answer

A point travels 10 units counterclockwise around the unit circle from (1, 0). How far past one full turn is it? Give a decimal to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

1301

QUESTION ID

S35-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$10 - 2\pi \approx 3.72$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 2

Read sine and cosine of radian angles from the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

What point of the unit circle is at angle π? Give cos π and sin π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

1303

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, 0); cos π = −1, sin π = 0

GRADING NOTE Form required: the point and both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find cos(3π/2) and sin(3π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

1304

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0 and −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

1305

QUESTION ID

S35-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Back at (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin(π/2) equals

A. −1

B. 0

C. π/2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

1306

QUESTION ID

S35-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE Point (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cos θ and sin θ make sense for θ = 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

1307

QUESTION ID

S35-F2-Q5

TYPE

Short answer

CORRECT ANSWER

13π/6 ends at the same point as π/6, so cos and sin take the values at that point (√3/2, 1/2).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Find exact trigonometric values beyond the first quadrant

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Special right triangle ratios	**HSG-SRT.C.6**
BACKWARD 2	Reference angles for radian measures	**HSF-TF.A.1**
BACKWARD 3	Signs of coordinates by quadrant	**6.NS.C.6.b**
FORWARD 1	Values for angles beyond 2π and solving for angles	**HSF-TF.A.2**
FORWARD 2	Verify the Pythagorean identity at exact values	**HSF-TF.C.8**

SET 36 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find sin(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

1310

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Form required: exact. Reference angle π/6, quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find cos(4π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

1311

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Form required: exact. Reference angle π/3, quadrant III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find tan(7π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

1312

QUESTION ID

S36-M-Q3

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Form required: exact. Point (√2/2, −√2/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find $\sin(-\pi/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

1313

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Form required: exact. Point (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the unit circle to find cos(11π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

1314

QUESTION ID

S36-M-Q5

TYPE

Short answer

CORRECT ANSWER

√3/2

GRADING NOTE Form required: exact. Reference angle π/6, quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 1

Special right triangle ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a 30°−60°−90° triangle with hypotenuse 1, give the side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2 (opposite 30°) and √3/2 (opposite 60°)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find sin 45° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

1317

QUESTION ID
S36-B1-Q2

TYPE
Short answer

CORRECT ANSWER
√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

True or false?

cos 60° = √3/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S36-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos 60° = 1/2; √3/2 is sin 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin 60° equals

A. √2/2

B. √3/2

C. √3

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

1319

QUESTION ID
S36-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. √3/2

GRADING NOTE Opposite/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find tan 30° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

1320

QUESTION ID
S36-B1-Q5

TYPE
Short answer

CORRECT ANSWER
√3/3

GRADING NOTE Any equivalent form (1/√3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 2

Reference angles for radian measures • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the reference angle of 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the reference angle of 4π/3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The reference angle of 7π/4 is 3π/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S36-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2\pi - 7\pi/4 = \pi/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which quadrant contains 11π/6?

A. I

B. IV

C. III

D. II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S36-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. IV

GRADING NOTE Between 3π/2 and 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 4π/3 to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

1326

QUESTION ID
S36-B2-Q5

TYPE
Short answer

CORRECT ANSWER
240°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q5**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 3

Signs of coordinates by quadrant • Grade 6 • The Number System • 6.NS.C.6.b

ANSWER KEY Reduced copy of the question page, with its answer

In which quadrant are both coordinates negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

1328

QUESTION ID

S36-B3-Q1

TYPE

Short answer

CORRECT ANSWER

III

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the signs of the coordinates in quadrant II?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

1329

QUESTION ID

S36-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x negative, y positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

True or false?

A point in quadrant IV has a positive y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

1330

QUESTION ID

S36-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Quadrant IV points have $x > 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

The point $(-\sqrt{3}/2, 1/2)$ is in quadrant

A. I

B. II

C. III

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

1331

QUESTION ID

S36-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. II

GRADING NOTE (−, +).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the signs of x and y for a point in quadrant IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

1332

QUESTION ID

S36-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x positive, y negative

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 1

Values for angles beyond 2π and solving for angles • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(13π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

1334

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find cos(−3π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

1335

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−√2/2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every θ with $0 \le \theta < 2\pi$ and $\cos\theta = -1/2$.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

1336

QUESTION ID

S36-F1-Q3

TYPE

Short answer

CORRECT ANSWER

2π/3 and 4π/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

tan(π/2) is undefined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

1337

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE cos(π/2) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which angle has sin θ = −√3/2 and cos θ = 1/2?

A. 5π/3

B. 4π/3

C. 2π/3

D. 7π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

1338

QUESTION ID

S36-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 5π/3

GRADING NOTE Quadrant IV, reference π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 2

Verify the Pythagorean identity at exact values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Verify $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 5\pi/6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

1340

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(1/2)^2 + (-\sqrt{3}/2)^2$
$= 1/4 + 3/4 = 1$

GRADING NOTE Form required: the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Verify $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 7\pi/4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

1341

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/2 + 1/2 = 1

GRADING NOTE Form required: the computation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin^2\theta + \cos^2\theta = 1$ is true only for first-quadrant angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

1342

QUESTION ID

S36-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every point (cos θ, sin θ) is on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If cos θ = −1/2 and θ is in quadrant III, then sin θ =

A. √3/2

B. −√3/2

C. 1/2

D. −1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

1343

QUESTION ID

S36-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −√3/2

GRADING NOTE $\sin^2 = 3/4$, negative in III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the values found from the unit circle always satisfy $\sin^2\theta + \cos^2\theta = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

1344

QUESTION ID

S36-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(cos θ, sin θ) lies on $x^2 + y^2 = 1$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Explain the extension of sine and cosine to all real numbers

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Right-triangle definitions of sine and cosine	**HSG-SRT.C.6**
BACKWARD 2	Points on the unit circle	**HSG-GPE.A.1**
BACKWARD 3	Radian measure as directed arc length	**HSF-TF.A.1**
FORWARD 1	Explain where tangent is defined and how values repeat	**HSF-TF.A.2**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 37 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which statement correctly explains how the unit circle defines cos t and sin t for every real number t?

A. Start at (0, 1) and travel t units clockwise; cos t is the y-coordinate and sin t is the x-coordinate of the endpoint.

B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.

C. Travel t units counterclockwise from (1, 0); for t < 0 the values are undefined.

D. Draw a right triangle with an acute angle of t radians; cos t and sin t are the ratios adjacent/hypotenuse and opposite/hypotenuse, so the definition works only for t between 0 and π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

1347

QUESTION ID

S37-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.

GRADING NOTE Arc length |t| from (1, 0), direction by the sign of t, coordinates of the endpoint. The others use the right-triangle definition (only acute angles), the wrong start and coordinates, or leave negative t undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which explanation shows why sin(t + 2π) = sin t for every real number t?

A. For t = π/2 both sides equal 1, so they are always equal.

B. Sine distributes over addition, so sin(t + 2π) = sin t + sin 2π, and sin 2π = 0 because the point at 2π is (1, 0); therefore sin(t + 2π) = sin t.

C. Adding 2π to t reflects the endpoint on the unit circle over the x-axis, and a reflection over the x-axis keeps the y-coordinate, so sin(t + 2π) = sin t.

D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and t + 2π end at the same point, which has the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

1348

QUESTION ID
S37-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and t + 2π end at the same point, which has the same y-coordinate.

GRADING NOTE One full turn returns to the same point. The others distribute sine over a sum (false in general), check one value, or describe the wrong motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

Why is cos t < 0 for π/2 < t < π?

A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

B. For these t the endpoint is in quadrant II, where y-coordinates are negative, and cos t is the y-coordinate.

C. cos(2π/3) = −1/2, so cos t is negative on this interval.

D. On the unit circle cos t decreases steadily as t increases from 0, so once t passes π/2, where cos t = 0, the value of cos t is negative for every larger t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

1349

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

GRADING NOTE Quadrant II and the x-coordinate. The others use the wrong coordinate and sign, a false claim (cos t increases again after π and is positive after 3π/2), or one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

Why is $-1 \le \sin t \le 1$ for every real number t?

A. $\sin(\pi/2) = 1$ and $\sin(3\pi/2) = -1$, so every value lies between them.

B. In a right triangle the hypotenuse is longer than either leg, so opposite/hypotenuse is less than 1, which means $\sin t < 1$ for every real number t.

C. sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

D. sin t is an arc length on the unit circle, and those arcs are at most 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

1350

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

GRADING NOTE Coordinates of points on the unit circle satisfy $x^2 + y^2 = 1$. The others only cover acute angles (and wrongly exclude sin t = 1), misstate the definition, or reason from two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

How does the unit-circle definition give sin(−π/6)?

A. Travel π/6 clockwise from (1, 0) to (1/2, −√3/2), so sin(−π/6) = −√3/2.

B. Negative inputs are not on the unit circle, so sin(−π/6) is undefined.

C. Travel π/6 clockwise from (1, 0) to the point (√3/2, −1/2), so sin(−π/6) = −1/2.

D. Travel π/6 counterclockwise from (1, 0) to the point (√3/2, 1/2), so sin(−π/6) = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

1351

QUESTION ID

S37-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Travel π/6 clockwise from (1, 0) to the point (√3/2, −1/2), so sin(−π/6) = −1/2.

GRADING NOTE A negative t means clockwise travel; the endpoint is (√3/2, −1/2) and the sine is its y-coordinate. The others go the wrong way, use the point for π/3, or leave negative t undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 1

Right-triangle definitions of sine and cosine • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Define sin A in a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

1353

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

opposite/hypotenuse

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a right triangle with hypotenuse 1, what are the legs next to and opposite an acute angle θ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

1354

QUESTION ID
S37-B1-Q2

TYPE
Short answer

CORRECT ANSWER
cos θ (adjacent) and sin θ (opposite)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

True or false?

The right-triangle definition works for θ = 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

1355

QUESTION ID

S37-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A right triangle has no 120° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

In a right triangle with hypotenuse 1, the leg opposite θ is

A. sin θ

B. tan θ

C. 1

D. cos θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

1356

QUESTION ID
S37-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. sin θ

GRADING NOTE opposite/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

Why does the right-triangle definition fail for negative angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

1357

QUESTION ID

S37-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Triangle angles are positive (between 0° and 90° for the acute angles).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 2

Points on the unit circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center (0, 0) and radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

1359

QUESTION ID
S37-B2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2 + y^2 = 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (−1/2, √3/2) on the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

1360

QUESTION ID
S37-B2-Q2

TYPE
Short answer

CORRECT ANSWER
Yes: 1/4 + 3/4 = 1

GRADING NOTE Form required: decision and check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

True or false?

Every point on the unit circle has $-1 \le x \le 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

1361

QUESTION ID

S37-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

Which point is on the unit circle?

A. (0.9, 0.9)

B. (0.5, 0.5)

C. (1, 1)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

1362

QUESTION ID
S37-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. (0.6, −0.8)

GRADING NOTE 0.36 + 0.64 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q4**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

A point on the unit circle has x = 0. Give both possible points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

1363

QUESTION ID
S37-B2-Q5

TYPE
Short answer

CORRECT ANSWER
(0, 1) and (0, −1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 3

Radian measure as directed arc length • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

What arc length on the unit circle does t = π/6 cut off?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

1365

QUESTION ID

S37-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

Which direction does t = −π/6 travel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

1366

QUESTION ID

S37-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

True or false?

t = π returns to the starting point (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

1367

QUESTION ID

S37-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE t = π is half a turn and ends at (−1, 0); t = 2π returns to (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc of length π from (1, 0) ends at

A. (0, 1)

B. (0, −1)

C. (1, 0)

D. (−1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

1368

QUESTION ID

S37-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (−1, 0)

GRADING NOTE Half a turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give a negative t that ends at (0, −1).

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

1369

QUESTION ID

S37-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−π/2

GRADING NOTE Any valid value (−π/2 + 2πk).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 1

Explain where tangent is defined and how values repeat • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Why is tan t undefined at t = π/2 + kπ (k an integer)?

A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

B. tan t is undefined whenever sin t = 1.

C. At these t the endpoint on the unit circle is (1, 0) or (−1, 0), so sin t = 0 and tan t = sin t/cos t would have to divide by 0.

D. tan t = cos t/sin t, and sin t = 0 at these t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

1371

QUESTION ID
S37-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

GRADING NOTE The endpoints are (0, ±1), where the x-coordinate cos t is 0. The others use the wrong points, invert the ratio, or give a rule that misses t = 3π/2 (where sin t = −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and range of y = sin t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

1372

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Domain all real numbers; range −1 ≤ y ≤ 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

cos t and sin t can both equal 0 for the same t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

1373

QUESTION ID

S37-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (0, 0) is not on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

The unit-circle definition agrees with the triangle definition for

A. $t < 0$ only

B. $\pi/2 < t < \pi$ only

C. all t

D. $0 < t < \pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

1374

QUESTION ID

S37-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $0 < t < \pi/2$

GRADING NOTE Acute angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cos t = cos(−t) using the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

1375

QUESTION ID

S37-F1-Q5

TYPE

Short answer

CORRECT ANSWER

t and −t end at points that are reflections over the x-axis, which have the same x-coordinate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle argument shows that sin(−t) = −sin t?

A. sin(−π/2) = −1 = −sin(π/2), so the rule holds for every t.

B. The endpoints for t and −t are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

C. The endpoints for t and −t are the same point, so their sines are opposites.

D. The endpoints for t and −t are always reflections of each other over the y-axis, so their y-coordinates (the sines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

1377

QUESTION ID
S37-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. The endpoints for t and −t are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

GRADING NOTE Equal arcs in opposite directions from (1, 0) give mirror images over the x-axis. The others name the wrong reflection (it keeps y), check one value, or describe the points wrongly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

Is sine an even function, an odd function, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

1378

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

tan(t + π) = tan t for every t where it is defined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

1379

QUESTION ID

S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The endpoints of t and t + π are opposite points; the ratio y/x is unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The period of y = tan t is

A. π

B. 4π

C. π/2

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

1380

QUESTION ID

S37-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. π

GRADING NOTE Opposite points give equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

Explain why cos is an even function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

1381

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

cos(−t) = cos t because reflection over the x-axis keeps the x-coordinate.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Write sinusoidal models with given amplitude, period, and midline

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Amplitude and midline from the maximum and minimum	**7.NS.A.3**
BACKWARD 2	Period and frequency	**6.RP.A.2**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Write models from data or with a phase shift	**HSF-TF.B.5**
FORWARD 2	Solve for times in sinusoidal models	**HSF-TF.B.7**

SET 38 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write a cosine function with amplitude 3, period 2π, midline y = 1, and a maximum at x = 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

1384

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

3cos x + 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 4 minutes. A rider starts at the bottom. Write the rider's height h after t minutes.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

1385

QUESTION ID
S38-M-Q2

TYPE
Short answer

CORRECT ANSWER
25 − 20cos(πt/2)

GRADING NOTE Any equivalent form. Amplitude 20, midline 25, period 4 so B = 2π/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Water depth at a harbor varies between 1 ft (low tide) and 9 ft (high tide) with a period of 12 hours. High tide is at t = 0. Write the depth d after t hours.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

1386

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

4cos(πt/6) + 5

GRADING NOTE Any equivalent form. Midline (9 + 1)/2, amplitude (9 − 1)/2, B = 2π/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A town's average temperature is lowest, 10°C, in month t = 0 and highest, 30°C, at t = 6, with a 12-month period. Write a model T(t).

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

1387

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

$20 - 10\cos(\pi t/6)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a sine function with amplitude 2, period π, and midline y = −1 that is at the midline and increasing at x = 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

1388

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

2sin(2x) − 1

GRADING NOTE Any equivalent form. B = 2π/π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 1

Amplitude and midline from the maximum and minimum • Grade 7 • The Number System • 7.NS.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find (9 + 1)/2 and (9 − 1)/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

1390

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 and 4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

A quantity varies between 10 and 30. Find the midline value and amplitude.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

1391

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Midline 20; amplitude 10

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

True or false?

The amplitude is the maximum minus the minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

1392

QUESTION ID

S38-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is half the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For max 45 and min 5, the midline is

A. y = 20

B. y = 50

C. y = 25

D. y = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

1393

QUESTION ID

S38-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. y = 25

GRADING NOTE (45 + 5)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m with center 25 m high. Give its maximum and minimum heights.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

1394

QUESTION ID

S38-B1-Q5

TYPE

Short answer

CORRECT ANSWER

45 m and 5 m

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 2

Period and frequency • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A wheel turns once every 4 minutes. How many turns per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

1396

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

A spring oscillates 3 times per second. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

1397

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/3 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

True or false?

Frequency is equal to the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

1398

QUESTION ID

S38-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Frequency is the reciprocal: f = 1/period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

A period of 12 hours means a frequency of

A. 12 cycles per hour

B. 2 cycles per hour

C. 1/2 cycle per hour

D. 1/12 cycle per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

1399

QUESTION ID

S38-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/12 cycle per hour

GRADING NOTE Reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

y = sin(Bx) has period 2π/B. Find B for period 4.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

1400

QUESTION ID

S38-B2-Q5

TYPE

Short answer

CORRECT ANSWER

π/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Describe how y = 3cos x + 1 is obtained from y = cos x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

1402

QUESTION ID

S38-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Vertical stretch by 3, then shift up 1

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe y = −cos x compared with y = cos x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

1403

QUESTION ID

S38-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Reflection over the x-axis

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

True or false?

y = cos x + 5 has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

1404

QUESTION ID

S38-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The maximum of cos x is 1, so the maximum is 6; 5 is the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

$y = -20\cos x$ starts (at $x = 0$) at

A. its minimum

B. its maximum

C. the midline going up

D. the midline going down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q4/5

1405

QUESTION ID

S38-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. its minimum

GRADING NOTE −20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write y = sin x shifted down 1 and stretched vertically by 2.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

1406

QUESTION ID

S38-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2sin x − 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 1

Write models from data or with a phase shift • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A rider's height is 3 m at t = 0 (bottom), 27 m at the top after 15 s, and the motion repeats every 30 s. Write h(t).

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

1408

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 − 12cos(πt/15)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

Sunrise times vary between 5:30 and 7:30 (as 5.5 and 7.5 hours) with period 365 days, latest at day 0. Write a model.

S = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

1409

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.5 + cos(2πt/365)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

True or false?

$y = 4\sin(x - \pi/2)$ is the same as $y = 4\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

1410

QUESTION ID

S38-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin(x - \pi/2) = -\cos x$, so $4\sin(x - \pi/2) = -4\cos x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

A sine model with period 10, amplitude 3, midline 7, and a maximum at t = 2.5 is

A. 7 + 3sin(10t)

B. 7 + 3sin(πt/5)

C. 3 + 7sin(πt/5)

D. 7 + 3cos(πt/5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

1411

QUESTION ID

S38-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7 + 3sin(πt/5)

GRADING NOTE Quarter period 2.5 gives the maximum of sine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model has a maximum of 18 at t = 3 and the next minimum of 6 at t = 7. Write a cosine model.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

1412

QUESTION ID

S38-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12 + 6cos(π(t − 3)/4)

GRADING NOTE Any equivalent form. Period 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 2

Solve for times in sinusoidal models • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

For h = 25 − 20cos(πt/2), find the first time t > 0 when the rider is at 35 m. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

1414

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4/3 min (about 1.33 min)

GRADING NOTE Any equivalent form. cos(πt/2) = −1/2 gives πt/2 = 2π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $d = 4\cos(\pi t/6) + 5$, find all times in $0 \le t < 12$ when the depth is 5 ft.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

1415

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 and 9 hours

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

True or false?

A sinusoidal model usually reaches a given value between its min and max twice per period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

1416

QUESTION ID

S38-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Once going up and once going down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For T = 20 − 10cos(πt/6), T = 25 first when

A. t = 2

B. t = 4

C. t = 3

D. t = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

1417

QUESTION ID

S38-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. t = 4

GRADING NOTE cos(πt/6) = −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

Explain why there are two solutions per period when the value is strictly between the minimum and maximum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

1418

QUESTION ID

S38-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The graph passes through that height once while rising and once while falling in each period.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

SET 39 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Interpret the parameters of sinusoidal models

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Build the model from amplitude, period, and midline	**HSF-TF.B.5**
BACKWARD 2	Evaluate a sinusoidal model	**HSF-TF.A.2**
BACKWARD 3	Convert rates between units	**6.RP.A.3.d**
FORWARD 1	Predict how changing a parameter changes the situation	**HSF-TF.B.5**
FORWARD 2	Interpret periodic features from graphs and tables	**HSF-IF.B.4**

SET 39 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ meters after t seconds. Give the maximum height, the minimum height, and the time for one revolution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

1421

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 m, 2 m, 30 s

GRADING NOTE Form required: all three with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Hours of daylight are $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$, t in days. Interpret 12, 3, and 365.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

1422

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

12: average daylight (midline, hours); 3: the most daylight varies from 12 (amplitude, hours); 365: days in one cycle (period)

GRADING NOTE Form required: all three interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

A model of blood pressure is P(t) = 100 + 20sin(2.5πt), t in seconds. Find the period and the number of heartbeats per minute.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

1423

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Period 0.8 s; 75 beats per minute

GRADING NOTE Form required: both. 2π/(2.5π) = 0.8; 60/0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

A spring's position is $y = -4\cos(3\pi t)$ cm, t in seconds. Give the amplitude, period, and frequency.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

1424

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 cm; 2/3 s; 1.5 cycles per second

GRADING NOTE Form required: all three with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two sounds are $y = \sin(880\pi t)$ and $y = \sin(440\pi t)$, t in seconds. Give the frequency of each in cycles per second and say which is higher.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

1425

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

440 and 220 cycles per second; the first is higher

GRADING NOTE Form required: both frequencies and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 1

Build the model from amplitude, period, and midline • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

What is the amplitude of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

1427

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the midline of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

1428

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

True or false?

The period of $y = \sin(Bt)$ is $B/(2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

1429

QUESTION ID

S39-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is $2\pi/B$ (for $B > 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

The period of $y = \sin(\pi t/15)$ is

A. 30

B. 15

C. $\pi/15$

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

1430

QUESTION ID

S39-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 30

GRADING NOTE $2\pi/(\pi/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a model with amplitude 4, period 2/3, and midline 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

1431

QUESTION ID

S39-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4sin(3πt)

GRADING NOTE Any equivalent form (cos also accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 2

Evaluate a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 12 + 10sin(π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

1433

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 12 + 10sin(3π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

1434

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-4\cos(0) = 0.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

1435

QUESTION ID

S39-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos 0 = 1, so −4cos 0 = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

100 + 20sin(π/2) equals

A. 100

B. 20

C. 120

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

1436

QUESTION ID

S39-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 120

GRADING NOTE sin(π/2) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 12 + 3sin(0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

1437

QUESTION ID

S39-B2-Q5

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 3

Convert rates between units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

A cycle lasts 0.8 s. How many cycles are in 60 s?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

1439

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

1.5 cycles per second is how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

1440

QUESTION ID

S39-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

True or false?

A period of 30 s means 30 revolutions per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

1441

QUESTION ID

S39-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 60/30 = 2 revolutions per minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

440 cycles per second is how many cycles per millisecond?

A. 4.4

B. 44

C. 440

D. 0.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

1442

QUESTION ID

S39-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.44

GRADING NOTE 1000 ms per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A tide has period 12 hours. How many cycles in 3 days?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

1443

QUESTION ID

S39-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 1

Predict how changing a parameter changes the situation • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel rider's height is h(t) = 12 + 10sin(πt/15) m after t seconds. The wheel is replaced by one of the same size that turns twice as fast. Rewrite h(t).

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

1445

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12 + 10sin(2πt/15)

GRADING NOTE Any equivalent form. Period 15 s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

Daylight in one city is D(t) = 12 + 3sin(2π(t − 80)/365) hours. A city farther from the equator has daylight between 8 and 16 hours with the same period and timing. Which parameter changes, and to what?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

1446

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The amplitude changes to 4

GRADING NOTE Form required: parameter and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

True or false?

Changing the midline changes the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

1447

QUESTION ID

S39-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The midline only shifts the graph vertically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

Multiplying B by 3 makes the period

A. 3 units longer

B. unchanged

C. three times as long

D. one third as long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

1448

QUESTION ID

S39-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. one third as long

GRADING NOTE 2π/(3B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A heart rate rises from 75 to 100 beats per minute. What is the new period, in seconds?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

1449

QUESTION ID
S39-F1-Q5

TYPE
Short answer

CORRECT ANSWER
0.6 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 2

Interpret periodic features from graphs and tables • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A table of a tide's depth shows maxima at t = 2 and t = 14 hours. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

1451

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12 hours

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

A graph of a temperature model has a high of 30 and a low of 10. What are the midline and amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

1452

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Midline 20; amplitude 10

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

True or false?

Consecutive minima of a periodic function are one period apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

1453

QUESTION ID

S39-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Repeating pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

On which interval does $h(t) = 12 + 10\sin(\pi t/15)$ increase?

A. $0 < t < 15$

B. $0 < t < 7.5$

C. $15 < t < 30$

D. $7.5 < t < 22.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

1454

QUESTION ID

S39-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 < t < 7.5

GRADING NOTE Rises to the maximum at t = 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

Explain how to read the frequency of a periodic graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

1455

QUESTION ID

S39-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Find the length of one cycle (the period) and take its reciprocal (cycles per unit).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Prove the Pythagorean identity

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	Coordinates of the point at angle θ	**HSF-TF.A.2**
BACKWARD 3	Expand and simplify algebraic squares	**HSA-SSE.A.2**
FORWARD 1	Prove identities derived from the Pythagorean identity	**HSF-TF.C.8**
FORWARD 2	Use the angle-addition formulas	**HSF-TF.C.9**

SET 40 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^2\theta + \cos^2\theta = 1$ for every angle θ, using the unit circle?

A. On the unit circle $\sin\theta + \cos\theta = 1$ for every angle θ, because the coordinates of a point add to the radius; squaring both sides gives $\sin^2\theta + \cos^2\theta = 1$.

B. The point at angle θ is $(\sin\theta, \cos\theta)$, so $\sin\theta + \cos\theta = 1$.

C. For $\theta = 30°$: $(1/2)^2 + (\sqrt{3}/2)^2 = 1$, so the identity holds.

D. The point at angle θ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

1458

QUESTION ID

S40-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. The point at angle θ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

GRADING NOTE The coordinates of the point satisfy the circle equation. The others check one angle or rely on the false statement $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

A right triangle has legs a and b, hypotenuse c, and acute angle θ opposite a. Which argument proves $\sin^2\theta + \cos^2\theta = 1$?

A. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a + b)^2/c^2$, and $(a + b)^2 = c^2$ by the Pythagorean Theorem, so the sum is 1.

B. $\sin\theta = b/c$ and $\cos\theta = a/c$, so $\sin\theta + \cos\theta = (a + b)/c = 1$.

C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

D. In a 3-4-5 triangle, $(3/5)^2 + (4/5)^2 = 1$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

1459

QUESTION ID

S40-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

GRADING NOTE Both ratios and $a^2 + b^2 = c^2$. The others swap the ratios and use $a + b = c$, square a sum incorrectly, or check one triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

Which argument proves $1 + \tan^2\theta = 1/\cos^2\theta$ wherever $\cos\theta \neq 0$?

A. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ to get $1 + \cos^2\theta/\sin^2\theta = 1/\sin^2\theta$; since $\cos^2\theta/\sin^2\theta = \tan^2\theta$, this gives $1 + \tan^2\theta = 1/\cos^2\theta$.

B. For $\theta = 0$: $1 + 0 = 1/1$, so the identity holds.

C. Square $\tan\theta = \sin\theta/\cos\theta$ to get $\tan^2\theta = \sin^2\theta/\cos^2\theta = 1/\cos^2\theta$.

D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

1460

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

GRADING NOTE Divide the Pythagorean identity by $\cos^2\theta$. The others divide by $\sin^2\theta$ and then treat $\cos^2\theta/\sin^2\theta$ as $\tan^2\theta$ (it is $1/\tan^2\theta$), replace $\sin^2\theta$ by 1, or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

θ is in quadrant II, and its point on the unit circle is (x, y) with $x < 0 < y$. Which argument proves $\sin^2\theta + \cos^2\theta = 1$ with a reference triangle?

A. The reference triangle has legs x and y, so $x + y = 1$, which means $\cos\theta + \sin\theta = 1$.

B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

C. The legs of the reference triangle are x and y, and x is negative in quadrant II, so $x^2 + y^2 = -1$ there, which means $\cos^2\theta + \sin^2\theta = -1$ instead of 1.

D. The identity was proved for acute angles, so it is true in quadrant II with no further work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

1461

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

GRADING NOTE Side lengths are $|x|$ and $|y|$; squares remove the signs. The others add the legs, mishandle the negative coordinate, or skip the argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$?

A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

B. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta = 1$, so the term $2\sin\theta\cos\theta$ is extra.

C. For $\theta = 0$: $(0 + 1)^2 = 1 + 0$, so the identity holds.

D. $(\sin\theta + \cos\theta)^2 = 1^2 = 1$ because $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

1462

QUESTION ID
S40-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

GRADING NOTE Expand the square, then use the Pythagorean identity. The others drop the middle term, check one angle, or use the false statement $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 3 and 4. Find the hypotenuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

1464

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

A right triangle has hypotenuse 1 and a leg 0.6. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

1465

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, $a^2 + b^2 = c^2$ where c is the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

1466

QUESTION ID

S40-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

Legs 5 and 12 give hypotenuse

A. 60

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

1467

QUESTION ID

S40-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13

GRADING NOTE √169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

Legs a and b and hypotenuse c: write c^2 in terms of a and b.

c^2 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

1468

QUESTION ID

S40-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 2

Coordinates of the point at angle θ • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give the coordinates of the point on the unit circle at angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

1470

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(cos θ, sin θ)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3) and sin(2π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

1471

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 and √3/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

True or false?

sin θ is the x-coordinate of the point at angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

1472

QUESTION ID

S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The point at angle π/4 is

A. (1/2, √3/2)

B. (1, 1)

C. (√3/2, 1/2)

D. (√2/2, √2/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

1473

QUESTION ID

S40-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (√2/2, √2/2)

GRADING NOTE 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the radius of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

1474

QUESTION ID

S40-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 3

Expand and simplify algebraic squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

1476

QUESTION ID

S40-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$a^2 + 2ab + b^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(a/c)^2 + (b/c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

1477

QUESTION ID

S40-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$(a^2 + b^2)/c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

True or false?

$\sin^2\theta$ means $\sin(\theta^2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

1478

QUESTION ID

S40-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin^2\theta$ means $(\sin\theta)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

Dividing $x^2 + y^2 = 1$ by x^2 gives

A. $1 + y^2/x^2 = 1/x^2$

B. $1 + y^2 = 1/x$

C. $y^2 = 1$

D. $x^2 + y^2 = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

1479

QUESTION ID

S40-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $1 + y^2/x^2 = 1/x^2$

GRADING NOTE Divide each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(s + c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

1480

QUESTION ID

S40-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$s^2 + 2sc + c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 1

Prove identities derived from the Pythagorean identity • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$?

A. Divide both sides by $\sin^2\theta - \cos^2\theta$ to get $\sin^2\theta + \cos^2\theta = 1$, which is the Pythagorean identity, so the identity is true.

B. $\sin^4\theta - \cos^4\theta = (\sin\theta - \cos\theta)^4$, which equals $\sin^2\theta - \cos^2\theta$.

C. For $\theta = \pi/4$ both sides are 0, so the identity holds.

D. $\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

1482

QUESTION ID

S40-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D.
$\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta.$

GRADING NOTE Factor as a difference of squares and use the Pythagorean identity. The others assume the identity (and divide by an expression that can be 0), use the false rule $a^4 - b^4 = (a - b)^4$, or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $(1 - \cos\theta)(1 + \cos\theta) = \sin^2\theta$?

A. For $\theta = \pi/2$: $(1 - 0)(1 + 0) = 1 = \sin^2(\pi/2)$, so the identity holds.

B. $(1 - \cos\theta)(1 + \cos\theta) = 1 - 2\cos\theta + \cos^2\theta$, and by the Pythagorean identity this equals $\sin^2\theta$.

C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

D. $(1 - \cos\theta)(1 + \cos\theta) = 1 + \cos^2\theta$, which equals $\sin^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

1483

QUESTION ID

S40-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

GRADING NOTE Multiply (difference of squares), then use the identity. The others multiply incorrectly or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin \theta + \cos \theta = 1$ for every θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

1484

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\theta = \pi/4$ gives $\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equals $1 - \sin^2\theta$?

A. $\tan^2\theta$

B. $1 - \cos^2\theta$

C. $\cos\theta$

D. $\cos^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

1485

QUESTION ID
S40-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $\cos^2\theta$

GRADING NOTE Rearranged identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$?

A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$:
$\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

B. Add $\sin^2\theta$ to both sides to get $\cos^2\theta = 2\cos^2\theta - 1 + \sin^2\theta$, which is always true.

C. For $\theta = 0$: $1 - 0 = 2 - 1$, so the identity holds.

D. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - 1 - \cos^2\theta = -1$, so the identity is false for every θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

1486

QUESTION ID
S40-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

GRADING NOTE Substitute $\sin^2\theta = 1 - \cos^2\theta$ and subtract the whole expression. The others drop the parentheses, check one angle, or end with an unproved statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 2

Use the angle-addition formulas • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use sin(a + b) = sin a cos b + cos a sin b to find sin(75°) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

1488

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(√6 + √2)/4

GRADING NOTE Any equivalent form. a = 45°, b = 30°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use cos(a + b) = cos a cos b − sin a sin b with a = b = θ to write cos 2θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

1489

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\cos^2\theta - \sin^2\theta$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

True or false?

$\cos 2\theta = 1 - 2\sin^2\theta$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

1490

QUESTION ID

S40-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use the Pythagorean identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

sin 2θ equals

A. $\sin^2\theta - \cos^2\theta$

B. 2sin θ

C. 2sin θ cos θ

D. sin θ + cos θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

1491

QUESTION ID

S40-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2sin θ cos θ

GRADING NOTE a = b = θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

Use $(\sin \theta + \cos \theta)^2 = 1 + 2\sin \theta \cos \theta$ to write $(\sin \theta + \cos \theta)^2$ in terms of $\sin 2\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

1492

QUESTION ID

S40-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$1 + \sin 2\theta$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q5**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

SET 41 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Use the Pythagorean identity to find trigonometric values

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	Signs of sine, cosine, and tangent by quadrant	**HSF-TF.A.2**
BACKWARD 2	Square roots of fractions	**8.EE.A.2**
BACKWARD 3	Solve $x^2 = c$	**HSA-REI.B.4.b**
FORWARD 1	Find values when the quadrant is not given, and combine values	**HSF-TF.C.8**
FORWARD 2	Find the angles with inverse trigonometric functions	**HSF-TF.B.7**

SET 41 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 3/5 and θ is in quadrant II. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

1495

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

cos θ = −4/5, tan θ = −3/4

GRADING NOTE Form required: both exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = −5/13 and θ is in quadrant III. Find sin θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

1496

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

sin θ = −12/13, tan θ = 12/5

GRADING NOTE Form required: both exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

tan θ = −2 and θ is in quadrant IV. Find sin θ and cos θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

1497

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin θ = −2√5/5, cos θ = √5/5

GRADING NOTE Form required: both exact (−2/√5 and 1/√5 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = −1/3 and θ is in quadrant IV. Find cos θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

1498

QUESTION ID

S41-M-Q4

TYPE

Short answer

CORRECT ANSWER

cos θ = 2√2/3, tan θ = −√2/4

GRADING NOTE Form required: both exact (−1/(2√2) accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = 0.6 and sin θ < 0. Find sin θ and tan θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

1499

QUESTION ID

S41-M-Q5

TYPE

Short answer

CORRECT ANSWER

sin θ = −0.8, tan θ = −4/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 1

Signs of sine, cosine, and tangent by quadrant • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In quadrant II, which of sin, cos, tan are positive?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

1501

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

sin only

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrant III, what is the sign of tan θ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

1502

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Positive

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In quadrant IV, $\cos\theta < 0$ and $\sin\theta > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

1503

QUESTION ID

S41-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Quadrant IV has $x > 0$ and $y < 0$, so $\cos\theta > 0$ and $\sin\theta < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $\sin\theta < 0$ and $\cos\theta > 0$, θ is in quadrant

A. I

B. II

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

1504

QUESTION ID

S41-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. IV

GRADING NOTE (+, −).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

If tan θ < 0 and sin θ > 0, which quadrant is θ in?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

1505

QUESTION ID

S41-B1-Q5

TYPE

Short answer

CORRECT ANSWER

II

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 2

Square roots of fractions • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find √(16/25).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

1507

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find √(8/9) in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

1508

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2√2/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

√(144/169) = 12/169.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

1509

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Take the root of both: √144/√169 = 12/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

√(1/5) equals

A. 5

B. √5

C. 1/25

D. √5/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

1510

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √5/5

GRADING NOTE Rationalize.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find √0.64.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

1511

QUESTION ID

S41-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 3

Solve $x^2 = c$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 16/25$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

1513

QUESTION ID

S41-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4/5 or −4/5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 1 - 9/25$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

1514

QUESTION ID

S41-B3-Q2

TYPE

Short answer

CORRECT ANSWER

±4/5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

True or false?

$x^2 = 0.36$ has only the solution 0.6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

1515

QUESTION ID

S41-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also −0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 + (3/5)^2 = 1$ gives x =

A. ±2/5

B. 2/5

C. 4/5

D. ±4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

1516

QUESTION ID

S41-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ±4/5

GRADING NOTE $x^2 = 16/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5x^2 = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

1517

QUESTION ID

S41-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$\pm\sqrt{5}/5$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 1

Find values when the quadrant is not given, and combine values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

ANSWER KEY Reduced copy of the question page, with its answer

sin θ = 3/5. List both possible values of cos θ and the quadrants they go with.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

1519

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4/5 (quadrant I) or −4/5 (quadrant II)

GRADING NOTE Form required: both with quadrants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$\cos\theta = -5/13$ and θ is in quadrant III. Find $\sin\theta \cdot \cos\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

1520

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60/169

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Knowing tan θ = 1 determines sin θ uniquely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

1521

QUESTION ID

S41-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE θ can be in quadrant I or III, so sin θ = ±√2/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $\sin\theta = 0.8$ and $\tan\theta < 0$, then $\cos\theta =$

A. -0.8

B. 0.2

C. 0.6

D. -0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

1522

QUESTION ID

S41-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. −0.6

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $1 + \tan^2\theta = 1/\cos^2\theta$ to find $\cos\theta$ if $\tan\theta = 3$ and θ is in quadrant III.

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

1523

QUESTION ID
S41-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$-\sqrt{10}/10$

GRADING NOTE Any equivalent form ($-1/\sqrt{10}$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 2

Find the angles with inverse trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

$\sin\theta = 3/5$ with $0 \le \theta < 2\pi$. Find both values of θ to the nearest thousandth. A calculator may be used.

$\theta \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

1525

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.644 and 2.498

GRADING NOTE Form required: both. arcsin(0.6) and π − arcsin(0.6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

cos θ = −5/13 and θ is in quadrant III. Find θ to the nearest thousandth (0 ≤ θ < 2π). A calculator may be used.

θ ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

1526

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4.32

GRADING NOTE Accept 4.318 to 4.320. 2π − arccos(−5/13).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

arcsin gives only angles between −π/2 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

1527

QUESTION ID

S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

The quadrant II angle with sin θ = 3/5 is

A. π + arcsin(3/5)

B. π − arcsin(3/5)

C. arcsin(3/5)

D. −arcsin(3/5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

1528

QUESTION ID

S41-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. π − arcsin(3/5)

GRADING NOTE Reflect over the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

Explain why the identity finds cos θ but the inverse function is needed to find θ itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

1529

QUESTION ID

S41-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The identity relates values to each other; finding the angle from a value requires undoing sine (with arcsin and the quadrant).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of polynomial and rational models

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret key features of linear and quadratic models	**HSF-IF.B.4**
BACKWARD 2	Describe increasing and decreasing behavior	**8.F.B.5**
BACKWARD 3	Zeros and sign from factors	**HSA-APR.B.3**
FORWARD 1	Sketch a model from a verbal description of key features	**HSF-IF.B.4**
FORWARD 2	Compare key features across representations	**HSF-IF.C.9**

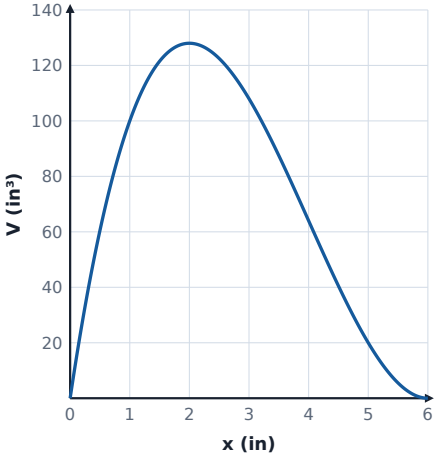
SET 42 • HSF-IF.B.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

An open box is made from a 12-in square sheet by cutting squares of side x from the corners. Its volume is $V(x) = x(12 - 2x)^2$. Use the graph to give the x that maximizes V, the maximum volume, and the interval where V increases.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

1532

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2 in; V = 128 in³; increases for 0 < x < 2

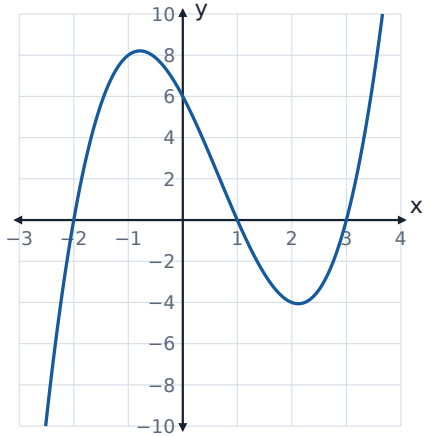
GRADING NOTE Form required: all three. Values read from the graph within ±0.2 in and ±5 in³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = (x + 2)(x - 1)(x - 3)$ is shown. Give the intervals where $f(x) > 0$ and the intervals where $f(x) < 0$.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

1533

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

$f > 0$ on $-2 < x < 1$ and $x > 3$; $f < 0$ on $x < -2$ and $1 < x < 3$

GRADING NOTE Form required: all intervals.

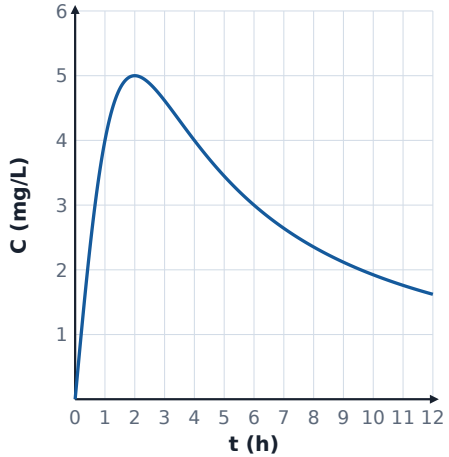
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Use the graph to give the maximum concentration, when it occurs, and the end behavior. Interpret the end behavior.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

1534

QUESTION ID
S42-M-Q3

TYPE
Short answer

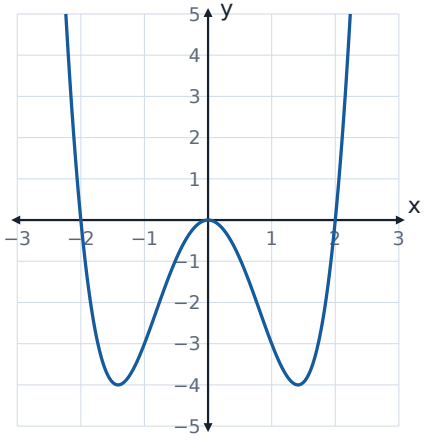
CORRECT ANSWER
Maximum 5 mg/L at $t = 2$ h; $C \to 0$ as t grows: the drug is eventually eliminated

GRADING NOTE Form required: maximum, time, end behavior and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

The graph of $f(x) = x^4 - 4x^2$ is shown. Give its x-intercepts and its relative maximum and minimum points.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

1535

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

x-intercepts −2, 0, 2; relative maximum (0, 0); relative minima at about (−1.41, −4) and (1.41, −4)

GRADING NOTE Form required: intercepts and all extrema (x within ±0.1).

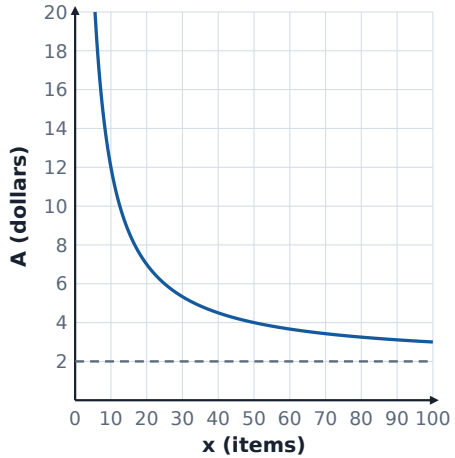
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The average cost of making x items is A(x) = (100 + 2x)/x dollars. Use the graph to describe the end behavior and interpret it.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

1536

QUESTION ID
S42-M-Q5

TYPE
Short answer

CORRECT ANSWER
As x grows, A(x) approaches $2 (horizontal asymptote y = 2): the average cost approaches the $2 cost of each item as the $100 fixed cost is spread out

GRADING NOTE Form required: the asymptote value and the interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 1

Interpret key features of linear and quadratic models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 32t + 4$. When is the maximum height, and what is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

1538

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t = 1; 20 ft

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What does the y-intercept of C = 25 + 3x mean if C is the cost of x rides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

1539

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A fixed cost of $25

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

True or false?

A parabola that opens down has a minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

1540

QUESTION ID

S42-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its vertex is the highest point: a maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 32t + 4$ increases for

A. all t

B. $0 < t < 1$

C. $t > 1$

D. $1 < t < 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

1541

QUESTION ID

S42-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 < t < 1

GRADING NOTE Before the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where is $y = (x - 2)(x - 6)$ negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

1542

QUESTION ID

S42-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$2 < x < 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 2

Describe increasing and decreasing behavior • Grade 8 • Functions • 8.F.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A graph rises from x = 0 to x = 2 and then falls.
Describe it in words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

1544

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Increasing on 0 < x < 2, decreasing after 2

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch-free: describe the graph of the concentration that rises then levels toward 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

1545

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Increases to a peak, then decreases and approaches 0

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

True or false?

A highest point between rising and falling parts is a relative maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5

1546

QUESTION ID

S42-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5

A function that falls on $1 < x < 3$ is

A. constant there

B. decreasing there

C. undefined there

D. increasing there

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

1547

QUESTION ID

S42-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. decreasing there

GRADING NOTE Falling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

What is a relative minimum?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

1548

QUESTION ID

S42-B2-Q5

TYPE

Short answer

CORRECT ANSWER

A point lower than all nearby points of the graph

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 3

Zeros and sign from factors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

1550

QUESTION ID

S42-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−2, 1, 3

GRADING NOTE Form required: all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

List the zeros of $x^4 - 4x^2 = x^2(x - 2)(x + 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

1551

QUESTION ID

S42-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0 (2), 2 (1), −2 (1)

GRADING NOTE Form required: all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q2**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

True or false?

$V(x) = x(12 - 2x)^2$ has zeros 0 and 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

1552

QUESTION ID
S42-B3-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $12 - 2x = 0$ at $x = 6$, so the zeros are 0 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$20t/(t^2 + 4) = 0$ when

A. never

B. $t = -2$

C. $t = 2$

D. $t = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

1553

QUESTION ID

S42-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. t = 0

GRADING NOTE Numerator 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $(x + 2)(x - 1)(x - 3)$ positive or negative at $x = 2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

1554

QUESTION ID

S42-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Negative: (4)(1)(−1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

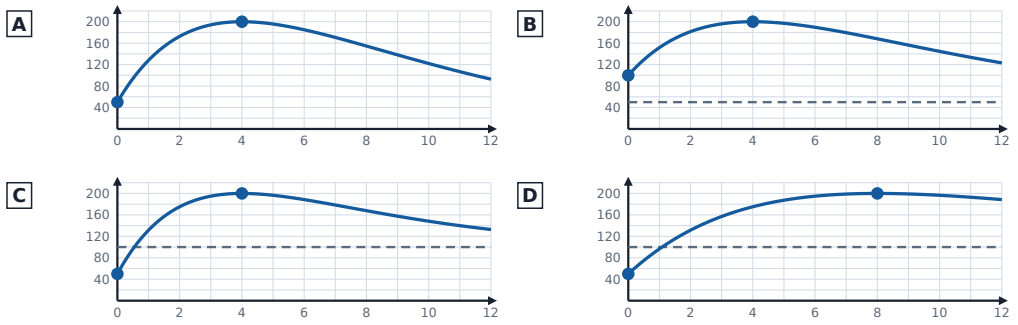
SET 42 • HSF-IF.B.4

FORWARD 1

Sketch a model from a verbal description of key features • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Which graph could show a population that is 50 at t = 0, rises to a maximum of 200 at t = 4, and then decreases toward a level of 100 as t grows?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

1556

QUESTION ID

S42-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE The graph must start at (0, 50), peak at (4, 200), and level off toward y = 100. The others level off toward 0, start at 100 and level off toward 50, or peak at t = 8.

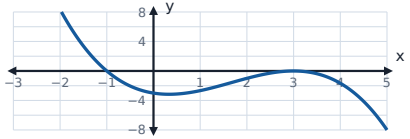
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

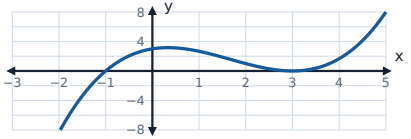
ANSWER KEY Reduced copy of the question page, with its answer

Which graph could show a cubic with zeros at −1 and 3 only, a negative y-intercept, falling to the right, and touching the x-axis at 3?

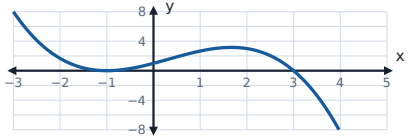
A



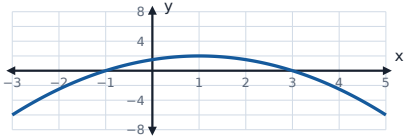
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

1557

QUESTION ID

S42-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Touching at 3 and falling to the right means the graph is below the axis on both sides of 3, so it is negative between −1 and 3 and the y-intercept is negative; for example $y = -(x + 1)(x - 3)^2/3$. The others rise to the right, touch at −1 instead of 3, or are a parabola (not a cubic).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

A graph can have a relative maximum that is not the absolute maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

1558

QUESTION ID
S42-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Cubics, for example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

A model rises to a peak and approaches 0 as t grows. Which formula fits?

A. $20t/(t^2 + 4)$

B. $20t$

C. t^2

D. $20 - t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

1559

QUESTION ID

S42-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $20t/(t^2 + 4)$

GRADING NOTE Rational with a peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a polynomial with zeros 0 and 6 and a relative maximum in between, whose values are positive between the zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

1560

QUESTION ID

S42-F1-Q5

TYPE

Short answer

CORRECT ANSWER

For example

$f(x) = x(6 - x)$

GRADING NOTE Any valid example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

FORWARD 2

Compare key features across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x(12 - 2x)^2$ and g is a quadratic with vertex (3, 120). Which has the greater maximum on $0 < x < 6$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

1562

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

f: 128 > 120

GRADING NOTE Form required: answer and values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$C(t) = 20t/(t^2 + 4)$ and a table of $D(t)$ gives $D(1) = 3$, $D(2) = 6$, $D(3) = 4$. Which has the greater peak among these values, and when?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

1563

QUESTION ID
S42-F2-Q2

TYPE
Short answer

CORRECT ANSWER
D: 6 at t = 2 (C peaks at 5)

GRADING NOTE Form required: function, value, time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

Two functions with the same zeros have the same relative extrema.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

1564

QUESTION ID
S42-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Scale changes the extrema values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

Which has a horizontal asymptote?

A. $(x + 2)(x - 1)(x - 3)$

B. $x^4 - 4x^2$

C. $x(12 - 2x)^2$

D. $(100 + 2x)/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

1565

QUESTION ID

S42-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (100 + 2x)/x

GRADING NOTE Rational with equal degrees.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to compare end behavior of a polynomial given by formula and a rational function given by a graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

1566

QUESTION ID
S42-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Use the leading term of the polynomial and read the rational graph's asymptote for large |x|.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret periodic models from graphs and tables

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret maximums and minimums on a graph	**HSF-IF.B.4**
BACKWARD 2	Read values from a table	**6.EE.C.9**
BACKWARD 3	Amplitude, period, and midline of a sinusoidal model	**HSF-TF.B.5**
FORWARD 1	Predict with periodicity	**HSF-IF.B.4**
FORWARD 2	Write a model from periodic features	**HSF-TF.B.5**

SET 43 • HSF-IF.B.4

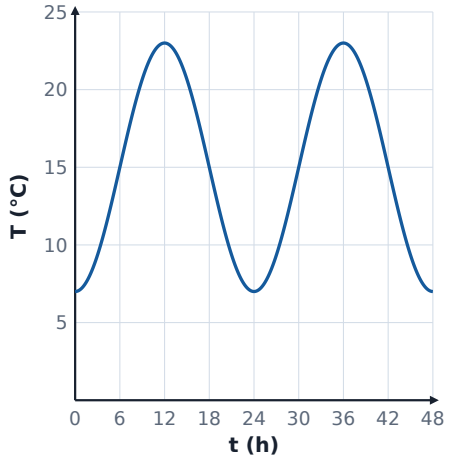
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a city's temperature T (°C) t hours after midnight on two days. Give the period, the maximum temperature, and the first time it occurs.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

1569

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

Period 24 h; maximum 23°C at t = 12 (noon)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

The table shows the water depth d (ft) at a harbor every 2 hours. Give the period, the depth at low tide, and the first time of low tide.

t (h)	0	2	4	6	8	10	12	14	16	18
d (ft)	9	7	3	1	3	7	9	7	3	1

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

1570

QUESTION ID
S43-M-Q2

TYPE
Short answer

CORRECT ANSWER
Period 12 h; low tide 1 ft at t = 6 h

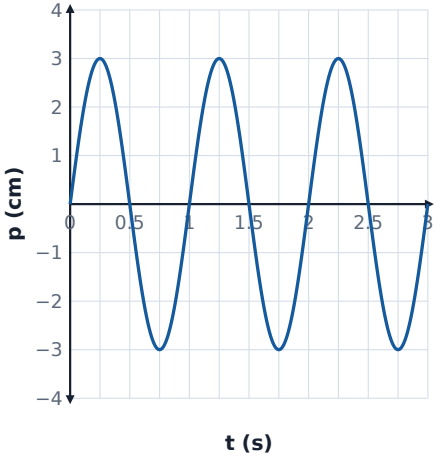
GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

A piston's position p (cm) is graphed for $0 \le t \le 3$ seconds. Give the amplitude, the period, and the intervals of the first cycle where p increases.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

1571

QUESTION ID

S43-M-Q3

TYPE

Short answer

CORRECT ANSWER

Amplitude 3 cm; period 1 s; increasing on $0 < t < 0.25$ and $0.75 < t < 1$

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

The table shows hours of daylight D in month m (m = 0 is January). Give the maximum, the minimum, the period, and the months in which daylight increases.

m	0	1	2	3	4	5	6	7	8	9	10	11	12
D (h)	9	9.4	10.5	12	13.5	14.6	15	14.6	13.5	12	10.5	9.4	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

1572

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

Maximum 15 h (m = 6), minimum 9 h (m = 0 and 12), period 12 months, increasing from m = 0 to m = 6

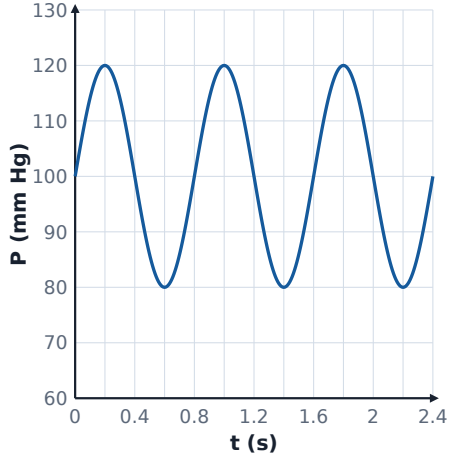
GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

The graph shows blood pressure P (mm Hg) over 2.4 seconds. Give the maximum and minimum pressures and the number of heartbeats per minute.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

1573

QUESTION ID

S43-M-Q5

TYPE

Short answer

CORRECT ANSWER

Maximum 120, minimum 80; period 0.8 s, so 75 beats per minute

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 1

Interpret maximums and minimums on a graph • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A graph of height peaks at 45 m. What does the peak mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

1575

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The greatest height reached is 45 m

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

A graph is lowest at (6, 1). Interpret it if it shows depth (ft) against time (h).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

1576

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The least depth, 1 ft, at 6 hours

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

True or false?

The minimum value of a graph is its lowest y-value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

1577

QUESTION ID

S43-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

A relative maximum on a graph is

A. the last point

B. any x-intercept

C. a point higher than nearby points

D. the y-intercept

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

1578

QUESTION ID

S43-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a point higher than nearby points

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

A graph rises on $0 < x < 2$. What does this mean for a temperature graph?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

1579

QUESTION ID

S43-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The temperature is increasing during that time

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 2

Read values from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. What is d when t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

1581

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 ft

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. At which listed times is d = 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

1582

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0 and 12

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

True or false?

A repeated pattern in a table suggests a periodic relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

1583

QUESTION ID

S43-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same values recur.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Hours of daylight in months m = 0, 3, 6, 12 are 9, 12, 15, 9. Daylight of 15 h occurs at

A. m = 12

B. m = 6

C. m = 0

D. m = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

1584

QUESTION ID

S43-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. m = 6

GRADING NOTE Read.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth is 7 ft at t = 2 h and 3 ft at t = 4 h. By how much does the depth d change from t = 2 to t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

1585

QUESTION ID

S43-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−4 ft

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 3

Amplitude, period, and midline of a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the amplitude and midline of $T = 15 - 8\cos(\pi t/12)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

1587

QUESTION ID

S43-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8 and y = 15

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of $p = 3\sin(2\pi t)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

1588

QUESTION ID

S43-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

True or false?

A wave between 80 and 120 has amplitude 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

1589

QUESTION ID

S43-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Amplitude is half the difference: (120 − 80)/2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

A wave with maximum 9 and minimum 1 has midline

A. y = 8

B. y = 5

C. y = 10

D. y = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

1590

QUESTION ID

S43-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 5

GRADING NOTE (9 + 1)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A period of 0.8 s means how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

1591

QUESTION ID

S43-B3-Q5

TYPE

Short answer

CORRECT ANSWER

75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 1

Predict with periodicity • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Use the temperature model $T(t) = 15 - 8\cos(\pi t/12)$ to find T at t = 100 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

1593

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

11°C

GRADING NOTE Any equivalent form. $100 = 4 \cdot 24 + 4$; $T(4) = 15 - 8\cos(\pi/3) = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depths repeat every 12 hours, and the depth is 1 ft at t = 6 h. Predict the depth at t = 30 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

1594

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1 ft

GRADING NOTE Any equivalent form. 30 = 24 + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

If f has period 12, then f(50) = f(10).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

1595

QUESTION ID

S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $50 = 4 \cdot 12 + 2$, so f(50) = f(2); f(10) need not equal it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

A piston's position is p = 3sin(2πt) cm, with period 1 s. At t = 7.25 s, p is

A. 0

B. 3

C. −3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

1596

QUESTION ID

S43-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE Same as t = 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Daylight follows a 12-month cycle with a maximum of 15 h at m = 6 (m = 0 is January). In which month after m = 12 does daylight next reach 15 h?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

1597

QUESTION ID

S43-F1-Q5

TYPE

Short answer

CORRECT ANSWER

m = 18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 2

Write a model from periodic features • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth varies between 9 ft (at t = 0) and 1 ft (at t = 6 h) and repeats every 12 hours. Write a cosine model for the depth d.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

1599

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 4cos(πt/6)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Daylight varies between 9 h (month m = 0) and 15 h (m = 6) and repeats every 12 months. Write a model for daylight D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5

1600

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$12 - 3\cos(\pi m/6)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

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